ICMA224

Linear Algebra B

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Chapter 1

General Vector Spaces

1.1 Vector Spaces

In Linear Algebra A, we discussed vectors in $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^4$. In it, we learned that one can view vectors both as algebraic and geometric objects. For example, consider the following vector in $\mathbb{R}^2$

$$\mathbf{v} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

What is $\mathbb{R}^2$? It is defined to be set of all vectors with two elements, i.e.,

$$\mathbb{R}^2 = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x, y \in \mathbb{R} \right\},$$

also know as *2-Euclidean Space*. We can define operations on this space, such as addition

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Observe that the sum of two vectors in $\mathbb{R}^2$ is also in $\mathbb{R}^2$, which is because the space is *closed* under addition. We can also perform scalar multiplication, e.g.,

$$5 \cdot \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 10 \\ 15 \end{pmatrix}.$$

Scalar multiplication is also a closed operation. We can generalize this idea using the concept of *vector spaces*[*], defined below.

Definition 1 (Vector Space). A vector space over $\mathbb{R}$ is a non-empty set V with two operations: vector addition and scalar multiplication. A vector space satisfies the following addition axioms:

A1) **Closure**: $\forall \mathbf{u}, \mathbf{v} \in V, \mathbf{u} + \mathbf{v} \in V$

A2) **Commutativity**: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

[*] No, not fields (yet).

A3) **Associativity**: $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$

A4) **Additive Identity**: $\exists \mathbf{0} \in V$ such that $\mathbf{u}+\mathbf{0}=\mathbf{u}$. This vector is known as the zero vector.

A5) **Additive Inverse**: $\forall \mathbf{v} \in V, \exists -\mathbf{v}$ such that $\mathbf{v}+(-\mathbf{v})=\mathbf{0}$.

It also satisfies the following scalar multiplication axioms:

S1) **Closure**: $\forall k \in \mathbb{R}, \forall \mathbf{v} \in V, k \cdot \mathbf{v} \in V$

S2) **Distributive (over vector sum)**: $k \cdot (\mathbf{u}+\mathbf{v}) = k\mathbf{u}+k\mathbf{v}$

S3) **Distributive (over scalar sum)**: $(k+m)\cdot \mathbf{v} = k \cdot \mathbf{v} + m \cdot \mathbf{v}$

S4) **Associativity**: $(km)\cdot \mathbf{v} = k \cdot (m \cdot \mathbf{v})$

S5) **Unity**: $1 \cdot \mathbf{v} = \mathbf{v}$.

Given a set with definitions for addition and scalar multiplication, one can verify that it is a vector space by verifying that it satisfies all ten of the aforementioned axioms.

1.1.1 Examples of Vector Spaces

The Euclidean Spaces

We define $\mathbb{R}^n$ to be the set of all n-tuples with entries in $\mathbb{R}$:

$$\mathbb{R}^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : x_1, \ldots, x_n \in \mathbb{R} \right\}$$

Vector and scalar multiplication by $k \in \mathbb{R}$ happens elementwise:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}, \quad k \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} kx_1 \\ \vdots \\ kx_n \end{pmatrix}$$

Therefore, $\mathbb{R}^n$ is a vector space over $\mathbb{R}$. It is called a *Euclidean Space*.

The Matrix Spaces

The vector space of all $m \times n$ matrices over the real numbers is also a vector space. It is defined as

$$M_{m\times n}(\mathbb{R}) = \left\{ \begin{pmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & & \\ a_{m1} & \ldots & a_{mn} \end{pmatrix} : a_{ij} \in \mathbb{R}, \forall i, j \right\}$$

We again define vector addition and scalar multiplication by $k \in \mathbb{R}$ elementwise.

The Complex Numbers

The set of *complex numbers* $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$ is also a vector space. We can define vector addition by

$$(a+bi)+(c+di) = (a+c)+(b+d)i$$

and scalar multiplication by

$$k(a+bi) = ka + kbi, \quad k \in \mathbb{R}.$$

With the operations above, $\mathbb{C}$ becomes a vector space over $\mathbb{R}$.

The Function Spaces

Let $[a,b]$ be a closed interval and $\mathbf{F}[a,b]$ be the set of all real-valued functions $f:[a,b]\to\mathbb{R}$. We can define vector addition and scalar multiplication by $k\in\mathbb{R}$ by

$$\begin{aligned}(f+g)(x)&=f(x)+g(x)\\(kf)(x)&=kf(x)\end{aligned}$$

Thus $\mathbf{F}[a,b]$ is a vector space over $\mathbb{R}$. This is the vector space of all real-valued functions defined on the interval $[a,b]$. Observe that we can still satisfy the axioms of a vector space. We can define the zero-function to be $\mathbf{0}(x)=0$ for all x and additive inverse function $(-f)(x)=-f(x)$.

The Polynomial Spaces

We define $\mathbf{P}_n(x)$ to be the set of all polynomials of degree at most n, i.e.,

$$\mathbf{P}_n(x)=\{a_0+a_1x+a_2x^2+\cdots+a_nx^n : a_0,a_1,\ldots,a_n\in\mathbb{R}\}.$$

The element with all $a_i=0$ is the zero polynomial $\mathbf{0}(x)=0^{\dagger}$. We can define vector addition and scalar multiplication by $k\in\mathbb{R}$ as follows: if $p(x)=\sum_{j=1}^n a_jx^j$ and $q(x)=\sum_{j=1}^n b_jx^j$, then

$$p(x)+q(x)=\sum_{j=1}^n(a_j+b_j)x^j.$$

Scalar multiplication is defined as

$$kp(x)=\sum_{j=1}^n(ka_j)x^j.$$

Then, $\mathbf{P}_n(x)$ is a vector space over $\mathbb{R}$.

A Non-Standard Vector Space

Let $V=\{(x_1,x_2):x_1,x_2\in\mathbb{R}\}$. Define vector addition by

$$\begin{pmatrix}x_1\\x_2\end{pmatrix}+\begin{pmatrix}y_1\\y_2\end{pmatrix}=\begin{pmatrix}x_1+y_1+1\\x_2+y_2+1\end{pmatrix}$$

and define scalar multiplication by $k\in\mathbb{R}$ to be

$$k\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}kx_1+k-1\\kx_2+k-1\end{pmatrix}$$

Let's consider an example

$$\begin{pmatrix}2\\3\end{pmatrix}+\begin{pmatrix}1\\5\end{pmatrix}=\begin{pmatrix}4\\9\end{pmatrix},\quad 4\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}8+4-1\\12+4-1\end{pmatrix}=\begin{pmatrix}11\\15\end{pmatrix}.$$

We claim that all vector space axioms are satisfied. Let's verify a few. Clearly the closure axioms is satisfied. How about the zero vector? We want a vector $\mathbf{0}=\begin{pmatrix}a & b\end{pmatrix}$ such that

$$\begin{pmatrix}a\\b\end{pmatrix}+\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x\\y\end{pmatrix},\quad\forall x,y.$$

†The degree of the zero polynomial is not defined.

Using the rules of addition for this vector space, we have that

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a+x+1 \\ b+y+1 \end{pmatrix}.$$

Therefore, our zero vector would be

$$\mathbf{0} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}.$$

Let's verify one more axiom – the distributive property over a vector sum. We have that for the left hand side

$$\begin{aligned} k \cdot \left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right] &= k \cdot \left[\begin{pmatrix} x_1 + y_1 + 1 \\ x_2 + y_2 + 1 \end{pmatrix} \right] \\ &= \begin{pmatrix} kx_1 + ky_1 + 2k - 1 \\ kx_2 + ky_2 + 2k - 1 \end{pmatrix}. \end{aligned}$$

Working backwards from the right hand side, we have

$$\begin{aligned} (k \cdot \mathbf{u}) + (k \cdot \mathbf{v}) &= \begin{pmatrix} kx_1 + k - 1 \\ kx_2 + k - 1 \end{pmatrix} + \begin{pmatrix} ky_1 + k - 1 \\ ky_2 + k - 1 \end{pmatrix} \\ &= \begin{pmatrix} kx_1 + ky_1 + 2k - 2 + 1 \\ kx_2 + ky_2 + 2k - 2 + 1 \end{pmatrix} \\ &= \begin{pmatrix} kx_1 + ky_1 + 2k - 1 \\ kx_2 + ky_2 + 2k - 1 \end{pmatrix}. \end{aligned}$$

Therefore, the S2 axiom is satisfied. We will not verify the rest since it's super tedious, however, V is indeed a vector space. Let's do another example of a non-standard vector space.

Example 1. Let V denote the set of all ordered pairs (x, y) and define addition in V as in $\mathbb{R}^2$. However, define a new scalar multiplication in V by

$$a \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ay \\ ax \end{pmatrix}.$$

Determine if V is a vector space with these operations. Observe that all the addition axioms are satisfied, as addition is defined the same as in $\mathbb{R}^2$. One easy axiom to check is unity (S5) – i.e., $1 \cdot \mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$. In this case

$$1 \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}.$$

Are these two matrices equal for all x and y? Obviously, they are not. We can easily show so with a counterexample. Therefore, as V does not satisfy the unity axiom, it is not a vector space.

1.1.2 Vector Space Properties

Next we will discuss a few small results, beginning with the fact that the zero vector is unique.

Lemma 1. In any vector space, the zero vector is unique.

Proof (LAETR). Let V be a vector space. Assume for the sake of contradiction that there exist two additive identities, $\mathbf{0}$ and $\mathbf{0}'$. By definition of the additive identity, we have that for any vector $\mathbf{u} \in V$

$$\mathbf{u} + \mathbf{0} = \mathbf{u} = \mathbf{u} + \mathbf{0}'.$$

Furthermore, we have that

$$\begin{aligned} \mathbf{0} &= \mathbf{0} + \mathbf{0}' \\ &= \mathbf{0}' + \mathbf{0} && \text{(Commutativity)} \\ &= \mathbf{0}'. && \text{(Additive identity)} \end{aligned}$$

Therefore, $\mathbf{0} = \mathbf{0}'$. □

The next is that vectors satisfy the property of *cancellation*.

Lemma 2 (Cancellation). If $\mathbf{u} + \mathbf{w} = \mathbf{v} + \mathbf{w}$, then $\mathbf{u} = \mathbf{v}$.

Proof.

$$\begin{aligned} \mathbf{u} + \mathbf{w} &= \mathbf{v} + \mathbf{w} \\ (\mathbf{u} + \mathbf{w}) + (-\mathbf{w}) &= (\mathbf{v} + \mathbf{w}) + (-\mathbf{w}) && \text{(Additive Inverse)} \\ \mathbf{u} + (\mathbf{w} + (-\mathbf{w})) &= \mathbf{v} + (\mathbf{w} + (-\mathbf{w})) && \text{(Associativity)} \\ \mathbf{u} + \mathbf{0} &= \mathbf{v} + \mathbf{0} \\ \mathbf{u} &= \mathbf{v} && \text{(Additive Identity)} \end{aligned}$$

□

The next result involves some smaller properties regarding the zero vector and scalar multiplication.

Lemma 3.

1) $0 \cdot \mathbf{v} = \mathbf{0}$
2) $(-1) \cdot \mathbf{v} = -\mathbf{v}$
3) $k \cdot \mathbf{0} = \mathbf{0}$
4) If $k \cdot \mathbf{v} = \mathbf{0}$ then either $k = 0$ or $\mathbf{v} = \mathbf{0}$
5) $(-k) \cdot \mathbf{v} = -(k \cdot \mathbf{v})$

Proof (LAETR). Let $\mathbf{v}$ be a vector in some vector space V and let $k \in \mathbb{R}$.

1) Due to the distributive property we have that

$$\begin{aligned} 0 \cdot \mathbf{v} &= (0 + 0) \cdot \mathbf{v} \\ &= 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v}. \end{aligned}$$

Adding the additive inverse to both sides, we have

$$\begin{aligned} 0 \cdot \mathbf{v} + (-0 \cdot \mathbf{v}) &= 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v} + (-0 \cdot \mathbf{v}) \\ \mathbf{0} &= 0 \cdot \mathbf{v} + \mathbf{0} \\ \mathbf{0} &= 0 \cdot \mathbf{v}, \end{aligned}$$

as desired.

2) This is equivalent to showing that
$$\mathbf{v} + -1 \cdot \mathbf{v} = \mathbf{0}.$$
We can do so as follows
$$\begin{aligned} \mathbf{v} + -1 \cdot \mathbf{v} &= (1 + (-1))\mathbf{v} \\ &= 0 \cdot \mathbf{v} \\ &= 0. && \text{(From 1)} \end{aligned}$$

3) Consider
$$\begin{aligned} \mathbf{0} + \mathbf{0} &= \mathbf{0} \\ k \cdot (\mathbf{0} + \mathbf{0}) &= k \cdot \mathbf{0} \\ k \cdot \mathbf{0} + k \cdot \mathbf{0} &= k \cdot \mathbf{0}. \end{aligned}$$
As we know the zero vector is unique, the only way this equation can be satisfied is to have
$$k \cdot \mathbf{0} = \mathbf{0}.$$

4) Follows from 1) and 3).

5) From 2) we know that the negative vector is equivalent to the additive inverse. Therefore, we can use the property of commutativity
$$\begin{aligned} -k \cdot \mathbf{v} &= (-1 \cdot k) \cdot \mathbf{v} \\ &= k \cdot (-\mathbf{v}) \\ &= -(k \cdot \mathbf{v}). \end{aligned}$$

□

1.2 Subspaces and Spanning Sets

1.2.1 Subspaces

Next we will introduce the concept of *subspaces* of a vector space.

Definition (Subspace). Let V be a vector space over $\mathbb{R}$ and let U be a nonempty subset of V. We say that U is a *subspace* of V, written $U \leq V$, if U is itself a vector space with the same operations in V.

For example, suppose V is a vector space and we define U to be a set containing the zero vector of V, i.e., $U = \{\mathbf{0}_V\}$. We can trivially verify that all ten axioms of a vector space are satisfied, and hence, $U \leq V$. In this instance, we would typically call U a *trivial subspace* of V. Another trivial subspace of V would be V itself. We can test to see if any subset of a vector space is a subspace using the *subspace test*.

Lemma 4 (Subspace Test). Let V be a vector space and let $U \subseteq V$, where $U \neq \emptyset$. Then U is a subspace of V if and only if

1) The zero vector exists in U.
2) U is closed under addition.
3) U is closed under scalar multiplication.

In most cases, we only need to verify the three conditions above and the rest of the axioms will follow automatically.

Example 2. Suppose $V = \mathbb{R}^2$ and $U = \{(x,0) : x \in \mathbb{R}\}$. Geometrically, this means that U is the set of all vectors along the x-axis. Clearly, the zero vector $(0,0) \in U$. For any two vectors in U, we have that

$$\begin{pmatrix} x \\ 0 \end{pmatrix} + \begin{pmatrix} y \\ 0 \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \end{pmatrix} \in U.$$

We also have that for $\lambda \in \mathbb{R}$

$$\lambda \begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} \lambda x \\ 0 \end{pmatrix} \in U.$$

Therefore, U is a subspace of $\mathbb{R}$.

We can also show the same is true for the vertical axis, and more generally, any single vector in $\mathbb{R}^2$. In fact, we can usually construct an infinite amount of subspaces for any vector space.

Example 3. Let $V = \mathbb{R}^2$ and let $U = \{\lambda(a,b) : \lambda \in \mathbb{R}\}$. U is the set of all possible vectors along the line (a,b). One can show that despite any operations performed, the result will always be on the line (a,b).

More generally, any vector that passes through the origin has an infinite amount of subspaces.

Example 4. Let $\mathbf{v}$ be any vector in a vector space V. We will show that

$$U = \{k\mathbf{v} : k \in \mathbb{R}\}$$

is a subspace of V. For the zero vector

$$\mathbf{0} \cdot \mathbf{v} = \mathbf{0} \in U.$$

Furthermore, to see if U is closed under addition

$$\lambda\mathbf{v} + \mu\mathbf{v} = (\lambda + \mu)\mathbf{v} \in U.$$

Finally, to verify if U is closed under scalar multiplication

$$k(\lambda\mathbf{v}) = (k\lambda)\mathbf{v} \in U.$$

Thus, we can conclude that $U \leq V$.

In the next two examples, we show two likely contenders for subspaces of $\mathbb{R}^2$ that fail to satisfy the Subspace Test.

Example 5. Let $V = \mathbb{R}^2$ and $U = \{(a,b) : a,b \in \mathbb{Q}\} \subseteq V$. We have that $\mathbf{0} \in U$ and $\mathbf{u}+\mathbf{v} \in U$ since 0 is rational and adding two rationals is also a rational. However, we can show that U is not

closed under scalar multiplication. Consider

$$\sqrt{2}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}\sqrt{2}\\\sqrt{2}\end{pmatrix} \notin U.$$

Although the previous candidate satisfied the property of having a zero vector and being closed under addition, it was not closed under scalar multiplication. The same is true for the next example.

Example 6. Consider $V = \mathbb{R}^2$ and $U = \{(x,y) : x \geq 0, y \geq 0\} \subseteq \mathbb{R}^2$. Again, U satisfies the first two conditions of the subspace test, however it fails for the third. Consider

$$-1\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}-1\\-1\end{pmatrix} \notin U.$$

Therefore U is not a subspace of V.

Let's check a few more subspaces.

Example 7. Which of the following are subspaces of $\mathbb{R}^2$?

(a) $U = \left\{\begin{pmatrix}a\\b\end{pmatrix} : a, b \geq 0\right\}$.

We have just shown in the previous example that U is not a subspace.

(b) $U = \left\{\begin{pmatrix}a\\1\end{pmatrix} : a \in \mathbb{R}\right\}$

This fails to be a subspace for multiple reasons, but most clearly, the zero vector is not in U.

(c) $U = \left\{c\begin{pmatrix}0\\1\end{pmatrix} : c \in \mathbb{R}\right\}$.

Yes, it is.

(d) $U = \left\{\begin{pmatrix}a\\b\end{pmatrix} : a^2 = b^2\right\}$.

We can show that U is not closed under addition. Consider

$$\begin{pmatrix}1\\-1\end{pmatrix} + \begin{pmatrix}-1\\-1\end{pmatrix} = \begin{pmatrix}0\\-2\end{pmatrix} \notin U.$$

Therefore, U is not a subspace of $\mathbb{R}^2$.

The next example highlights a technique to apply the subspace test to a vectors satisfying a linear homogeneous equation in a generalized manner.

Example 8. Show that U is a subspace of $\mathbb{R}^3$

$$U = \left\{\begin{pmatrix}a\\b\\c\end{pmatrix} : 2a - 3b + 5c = 0\right\}$$

Is U closed under addition? Observe that we can rewrite this linear equation in $A\mathbf{x} = \mathbf{0}$ form.

$$\begin{pmatrix} 2 & -3 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \end{pmatrix}$$

Let $\mathbf{x}, \mathbf{y} \in U$. We have that

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

For scalar multiplication, we can do something similar. Let $\mathbf{x} \in U$ and $\lambda \in \mathbb{R}$. We have that

$$A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda\mathbf{0} = \mathbf{0}.$$

Since U satisfies all three conditions of the subspace test, it is a subspace of $\mathbb{R}^3$.

Let's generalize the result we just obtained in Example 26.

Lemma 5. The set of all solutions of a homogeneous linear system with n variables is a subspace of $\mathbb{R}^n$. In terms of notation

$$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} \leq \mathbb{R}^n$$

What this result basically states is that if we are given a set of vectors satisfying linear homogeneous equations, it automatically follows that it is a subspace of $\mathbb{R}^n$. Another useful lemma is the following, which states that the intersection of two subspaces is also a subspace.

Lemma 6. Let V be a vector space and let U, W be subspaces of V. Then

$$U \cap W = \{\mathbf{v} \in V : \mathbf{v} \in U \wedge \mathbf{v} \in W\} \leq V.$$

Let's work with an example in $\mathbb{R}^3$.

Example 9. Let $V = \mathbb{R}^3$. Any plane through the origin is a subspace. Why? The standard equation of any plane through the origin is $ax + by + cz = 0$, so the plane becomes a solution of a homogeneous linear system, so by Lemma 5 it is a subspace. Any intersection of planes through the origin is also a subspace by Lemma 6.

Let's now look at a past exam question.

Example 10. Which of the following are subspaces of $\mathbb{R}^3$?

(i) $U = \{(x, y, z) \mid 3x - 2y + 4z = 0\}$

This is a subspace, by lemma 5.

(ii) $U = \{(x, y, z) \mid xz = 0\}$

This is not closed under addition. Consider

$$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \notin U$$

(iii) $U = \{(x, y, z) \mid 5y = 3z\}$

This is a subspace as we can rewrite as a homogeneous linear equation $5y - 3z = 0$, then apply lemma 5.

(iv) $U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \frac{x}{5} = \frac{y}{3} = 7z \right\}$

This is secretly a set of linear equations in disguise

$$\begin{aligned} \frac{x}{5} = \frac{y}{3} &\iff 3x - 5y = 0 \\ \frac{x}{5} = 7z &\iff x - 35z = 0 \\ \frac{y}{3} = yz &\iff y - 21z = 0 \end{aligned}$$

As this is clearly a system of linear homogeneous equations, U is also a subspace by lemma 5.

(v) $\mathbb{R}^2$.

$\mathbb{R}^2$ is not a *subset* of $\mathbb{R}^3$, therefore it cannot be a subspace.

The correct choices are (i), (iii), and (iv).

1.2.2 Spanning Sets

Recall that a *linear combination* of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is an expression of the form

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_n \mathbf{v}_n, \quad \lambda_i \in \mathbb{R}.$$

With this in mind we can define *spanning sets*.

Definition (Span). Let V be a vector space. Let $S \subset V$. The *span* of S, denoted $\mathrm{Span}(S)$ is a set of all linear combinations of vectors in S.

As an example, consider $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\} \subset \mathbb{R}^2$. We have that

$$\mathrm{Span}(S) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda \in \mathbb{R} \right\}.$$

What if we have two vectors? Consider $S = \{(0,1), (1,0)\}$. The span is then

$$\mathrm{Span}(S) = \left\{ y \begin{pmatrix} 0 \\ 1 \end{pmatrix} + x \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \right\} = \mathbb{R}^2.$$

Let's expand to $\mathbb{R}^3$. Suppose $S = \{\mathbf{v}\}$. Then $\mathrm{Span}(S)$ is a line in the direction of $\mathbf{v}$. What if $S = \{\mathbf{u}, \mathbf{v}\}$, where $\mathbf{u} \nparallel \mathbf{v}$? Then $\mathrm{Span}(S)$ is a plane in $\mathbb{R}^3$ containing $\mathbf{u}$ and $\mathbf{v}$.

Definition (Spanning Set). Suppose W is a vector space and $W = \mathrm{Span}(S)$. Then S is called a *spanning set* of W.

Take for example $\mathbb{R}^2$. A trivial spanning set of $\mathbb{R}^2$ is $\mathbb{R}^2$ itself. A more interesting one was the example we had earlier: $\{(0,1),(1,0)\}$. Something like $\{(0,2),(3,0)\}$ also works, we just have to play with the coefficients. It should be clear by now that spanning sets *are not unique*. In most cases, we want to find the smallest possible spanning set.

Example 11. Let $V = M_{2\times 2}(\mathbb{R})$. Any 2×2 matrix can be decomposed as the following linear combination

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore, the set

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

spans V.

We can also consider the polynomial spaces.

Example 12. Consider $V = \mathbf{P}_n(x) = \{a_0 + a_1 x + \cdots + a_n x^n : a_i \in \mathbb{R}\}$. The set $S = \{1, x, x^2, \ldots, x^n\}$ spans $\mathbf{P}_n(x)$.

Let's do an example with more computation.

Example 13. Let $\mathbf{x} = \begin{pmatrix} 2 & -1 & 2 & 1 \end{pmatrix}^T$ and $\mathbf{y} = \begin{pmatrix} 3 & 4 & -1 & 1 \end{pmatrix}^T$. Let $U = \text{Span}\{\mathbf{x}, \mathbf{y}\}$.

(a) Determine if the vector $\mathbf{p}$ is in U

$$\mathbf{p} = \begin{pmatrix} 0 \\ -11 \\ 8 \\ 1 \end{pmatrix}.$$

(b) Determine if the vector $\mathbf{q}$ is in U

$$\mathbf{q} = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 2 \end{pmatrix}.$$

For (a), we have that $\mathbf{p} \in U$ if and only if $\mathbf{p}$ is a linear combination of $\mathbf{x}$ and $\mathbf{y}$. In other terms, if there exists coefficients a and b that solve the equation

$$a\begin{pmatrix} 2 \\ -1 \\ 2 \\ 1 \end{pmatrix} + b\begin{pmatrix} 3 \\ 4 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -11 \\ 8 \\ 1 \end{pmatrix} \iff \begin{pmatrix} 2 & 3 \\ -1 & 4 \\ 2 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ -11 \\ 8 \\ 1 \end{pmatrix}.$$

In order to solve this, we would use Gaussian elimination to reduce to reduced row echelon form. Upon doing so, we obtain the array

$$\left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

Therefore, there exists the solution $(a,b) = (3,-2)$, and $\mathbf{p} \in U$. We can perform the same computation for $\mathbf{q}$ to obtain the reduced row echelon form

$$\left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array}\right)$$

As we have a bad row in row three, that means that no solution exists, i.e., it is not possible to write $\mathbf{q}$ as a linear combination of $\mathbf{x}$ and $\mathbf{y}$. In conclusion, $\mathbf{q}$ is not in U.

Next we will introduce another important result.

Theorem. Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\} \subset V$. Let $U = \text{Span}(S)$. Then

1) U is a subspace of V.
2) If W is subspace of V containing S, then $U \subseteq W$. In other terms, U is the smallest subspace of V that contains S.

Let's do an example of finding the spanning set for a vector space.

Example 14. Find a spanning set of

$$W = \left\{ \begin{pmatrix} a \\ a+b \\ 2b \\ b-a \end{pmatrix} : a, b \in \mathbb{R} \right\} \subseteq \mathbb{R}^4.$$

We can decompose this into

$$\begin{aligned} \begin{pmatrix} a \\ a+b \\ 2b \\ b-a \end{pmatrix} &= \begin{pmatrix} a \\ a \\ 0 \\ -a \end{pmatrix} + \begin{pmatrix} 0 \\ b \\ 2b \\ b \end{pmatrix} \\ &= a \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 2 \\ 1 \end{pmatrix}, \end{aligned}$$

which is a linear combination of a and b. Therefore, the set

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \\ 1 \end{pmatrix} \right\}$$

spans W.

1.3 Linear Independence and Basis

1.3.1 Linear Independence

Next we will introduce the concept of *linear independence*. Suppose we have three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in some vector space V and that $\mathbf{w}$ is some linear combination of $\mathbf{u}$ and $\mathbf{v}$, say $\mathbf{w} = \mathbf{u} + 2\mathbf{v}$. We can write any linear combination of $\mathbf{u}, \mathbf{v}, \mathbf{w}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$. For example

$$\begin{aligned} 3\mathbf{u} - \mathbf{v} + 2\mathbf{w} &= 3\mathbf{u} - \mathbf{v} + 2(\mathbf{u} + 2\mathbf{v}) \\ &= 5\mathbf{u} + 3\mathbf{v}. \end{aligned}$$

This implies that $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{Span}\{\mathbf{u}, \mathbf{v}\}$. This is ideal as we prefer spanning sets that are as small as possible. Let's introduce the formal definition for linear (in)dependence.

Definition (Linear Dependence). $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is *linearly dependent* if there exists a non-trivial linear combination that gives $\mathbf{0}$, i.e.,

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0},$$

where at least one $a_i \neq 0$. If S is not linearly dependent, we say it is *linearly independent*. In other words, whenever

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0},$$

then we must have $a_i = 0$, for all i.

Let's illustrate this idea with an example.

Example 15. Consider the set

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 0 \end{pmatrix} \right\} \subset \mathbb{R}^2.$$

Is it linearly dependent? We want to find a and b satisfying

$$a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Take $a = -5, b = 1$ as a non-trivial solution. Therefore, S is linearly dependent.

In the previous example, we saw that one vector was a linear combination of the other. This actually implies linear dependence in general.

Lemma 7. If $\exists v_i \in S$ that is a linear combination of the other vectors in S, then S is linearly dependent.

Let's do another example.

Example 16. Is the set

$$S = \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} \right\},$$

linearly independent? If we can find a non-trivial solution to

$$a\begin{pmatrix}-1\\0\\1\end{pmatrix}+b\begin{pmatrix}1\\1\\1\end{pmatrix}+c\begin{pmatrix}1\\3\\5\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix},$$

then S is linearly dependent. We can rewrite in $A\mathbf{x}=\mathbf{0}$ form

$$\begin{pmatrix}-1&1&1\\0&1&3\\1&1&5\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}.$$

We then form the augmented matrix and reduce to row echelon form

$$\left(\begin{array}{ccc|c}-1&1&1&0\\0&1&3&0\\1&1&5&0\end{array}\right)\underbrace{\rightsquigarrow}_{\text{Sage magic}}\left(\begin{array}{ccc|c}1&0&2&0\\0&1&3&0\\0&0&0&0\end{array}\right).$$

We see that there exists a solution to the system of equations. Most notably, we see that c is a free variable. This implies that there are infinitely many solutions of the form

$$\begin{aligned}a&=-2c\\b&=-3c\\c&=c.\end{aligned}$$

Therefore, we can conclude that S is linearly dependent.

In general, if we have k vectors

$$\mathbf{v}_1,\dots,\mathbf{v}_k\in\mathbb{R}^n,$$

we can determine if they are linearly (in)dependent by forming the linear combination

$$a_1\mathbf{v}_1+\cdots+a_k\mathbf{v}_k=\mathbf{0},$$

and the equation

$$\left(\begin{array}{c|c|c|c}\mathbf{v}_1&\mathbf{v}_2&\dots&\mathbf{v}_n\end{array}\right)\begin{pmatrix}a_1\\\vdots\\a_n\end{pmatrix}=\begin{pmatrix}0\\0\\\vdots\\0\end{pmatrix}.$$

Lemma 8. The columns of A are linearly independent if and only if the homogeneous linear system $A\mathbf{x}=\mathbf{0}$ has only the trivial solution.

The following example illustrates an application of lemma 8.

Example 17. Is the set $T=\{3x^2-x+2,x^2+x-1,x^2-3x+4\}$ linearly independent? T is linearly dependent if we can find coefficients a,b,c satisfying

$$a(3x^2-x+2)+b(x^2+x-1)+c(x^2-3x+4)=0.$$

Rewriting this in terms of each x term, we have

$$(3a+b+c)x^2+(-a+b-3c)x+(2a-b+4c)=0.$$

If we can show that this linear system only has the trivial solution, then we can immediately conclude that it is linearly independent. For that to be true, we require all coefficients of x^i to be zero. We have the system

$$\begin{aligned} 3a+b+c&=0\\ -a+b-3c&=0\\ 2a-b+4c&=0. \end{aligned}$$

We can form the augmented matrix and reduce it

$$\left(\begin{array}{ccc|c} 3 & 1 & 1 & 0\\ -1 & 1 & -3 & 0\\ 2 & -1 & 4 & 0 \end{array}\right) \underbrace{\rightsquigarrow}_{\text{Sage magic}} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0\\ 0 & 1 & -2 & 0\\ 0 & 0 & 0 & 0 \end{array}\right).$$

As we again have a free variable, c, there are infinitely many solutions. Therefore, the set T must be *linearly dependent*.

A more abstract problem is the next example.

Example 18. Given that $\{\mathbf{u},\mathbf{v}\}$ is linearly independent, show that $\{\mathbf{u}+\mathbf{v},\mathbf{u}-\mathbf{v}\}$ is also linearly independent. Let's consider

$$a(\mathbf{u}+\mathbf{v})+b(\mathbf{u}-\mathbf{v})=\mathbf{0}.$$

If we rewrite by collecting terms, we have

$$(a+b)\mathbf{u}+(a-b)\mathbf{v}=0.$$

This is now a linear combination of $\mathbf{u}$ and $\mathbf{v}$. Since $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, this implies that the coefficients of $\mathbf{u}$ and $\mathbf{v}$ must be zero, by definition of linear independence. In other terms

$$\begin{aligned} a+b&=0\\ a-b&=0, \end{aligned}$$

i.e., $a=0$ and $b=0$. Thus, $\mathbf{u}+\mathbf{v}$ and $\mathbf{u}-\mathbf{v}$ are linearly independent.

Next we will introduce a very useful theorem for reducing a spanning set.

Theorem (Plus/Minus Theorem). Let S be a nonempty subset of some vector space V. Let $W=\operatorname{Span}\{S\}$.

(i) If S is linearly independent and $\mathbf{v}\notin W$, then $S\cup\{\mathbf{v}\}$ is linearly independent.

(ii) If $\mathbf{v}\in S$ is a linear combination of other vectors in S, then $S\setminus\{\mathbf{v}\}$ still spans W.

Let's show this theorem with an example.

Example 19. Consider $V=\mathbb{R}^3$ and $S=\left\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$. Consider $\mathbf{v}=\begin{pmatrix}0\\0\\1\end{pmatrix}$. Then $S\cup\{\mathbf{v}\}$ is linearly

independent. If we have $S = \left\{ \left(\begin{smallmatrix}1\\0\\0\end{smallmatrix}\right), \left(\begin{smallmatrix}0\\1\\0\end{smallmatrix}\right), \left(\begin{smallmatrix}1\\1\\0\end{smallmatrix}\right) \right\}$, then

$$\text{Span}\{S\} = \text{Span}\left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix} \right\}$$

For more practice we will do another multi-part example.

Example 20.

(a) Consider

$$\left\{ \begin{pmatrix}2\\1\end{pmatrix}, \begin{pmatrix}-6\\-3\end{pmatrix} \right\} \subset \mathbb{R}^2.$$

Since the second vector is a linear combination of the first, we can throw it away without losing the span.

(b) Consider

$$\left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}4\\5\\0\end{pmatrix} \right\} \subset \mathbb{R}^2.$$

We can throw away the third vector as it is simply a linear combination of the first two.

(c) Consider

$$\{1, x, 2-x\} \subset \mathbf{P}_2(x).$$

We can again remove the third vector, since it is a linear combination of the first two:

$$2 - x = 2 \cdot x + -1 \cdot x$$

1.3.2 Basis

Definition (Basis). A set $S \subset V$ is called a *basis* for V if

1) S is linearly independent;
2) $\text{Span}(S) = V$.

Let's illustrate the concept of basis with an example.

Example 21. Consider $V = \mathbb{R}^2$ and $B = \{\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right), \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)\}$. We see that as B is linearly independent and that

$$\begin{pmatrix}x\\y\end{pmatrix} = x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}0\\1\end{pmatrix},$$

B is a basis for $\mathbb{R}^2$. Note that *basis is not unique*. Alternatively, if $B = \{\left(\begin{smallmatrix}2\\0\end{smallmatrix}\right), \left(\begin{smallmatrix}0\\3\end{smallmatrix}\right)\}$, we see that B is also linearly independent and

$$\begin{pmatrix}x\\y\end{pmatrix} = \frac{x}{2}\begin{pmatrix}2\\0\end{pmatrix} + \frac{y}{3}\begin{pmatrix}0\\3\end{pmatrix}.$$

Thus, B is also a basis for $V = \mathbb{R}^2$.

We also have the notion of *standard basis*, which is a basis that has a single nonzero entry with value 1.

- $\{\vec{e}_i, \dots, \vec{e}_n\}$ where $\vec{e}_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ 0 \end{pmatrix}$ is a *standard basis* for $\mathbb{R}^n$.
- $\{1, x, x^2, \dots, x^n\}$ is a standard basis for $\mathbf{P}_n(x)$.
- $\{E_{ij}, \forall i, j\}$ where E_{ij} is a zero matrix with entry (ij) equal to one is a standard basis for $M_{m\times n}(\mathbb{R})$.

The next result states that any two bases of a vector space have the same cardinality.

Theorem (Invariance Theorem). Let $B_1 = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $B_2 = \{\mathbf{u}, \dots, \mathbf{u}_n\}$ be bases for some vector space V. Then $m = n$. In other words, $|B_1| = |B_2|$.

The number of elements in the basis is known as the *dimension* of the vector space.

Definition (Dimension). Let V be a vector space and let B be a basis of V. $|B|$ is called the *dimension* of V denoted

$$\dim(V).$$

Some common dimensions include $\dim(\mathbb{R}^n) = n$, $\dim(\mathbf{P}_n(x)) = n + 1$, and $\dim(M_{m\times n}) = mn$. We note that the dimension of a set containing only the zero vector is zero.

Remark 1. $\dim\{\mathbf{0}\} := 0$.

A vector space V is *finite dimensional* if $\dim(V) < \infty$. What might be an example of an *infinite dimensional* vector space? One example might be $\mathbf{P}(x) = \{1, x, x^2, \dots\}$, the vector space of all polynomials. In this instance, we would have $\dim(\mathbf{P}(x)) = \infty$. The next theorem is very important for determining whether or not a subset of a vector space is linearly dependent.

Theorem. Suppose $\dim(V) = n > 0$. Let $S \subset V$.

i) If $|S| > n$, then S is linearly dependent.

ii) If $|S| < n$, then S does not span V.

Note that just because a set $S \subset V$ has cardinality $n = \dim(V)$, it does not automatically imply that S is a basis. We cannot make any conclusions. This is illustrated in the example below.

Example 22. Consider $V = \mathbb{R}^2$.

- If $|S| = 3$, then S is not linearly independent.
- If $|S| = 1$, then S does not span $\mathbb{R}^2$.
- If $|S| = 2$, then we cannot make any conclusions.
  - $S = \{\binom{1}{0}\}, \binom{2}{0}\}$ does not span $\mathbb{R}^2$ and is not linearly independent despite having $|S| = 2$.

Let's do another example.

Example 23. Consider the subspace $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subset \mathbb{R}^3$ where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Let's determine a basis for W. Clearly, $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ already spans W, as indicated by the definition of W. Is $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ also linearly independent? Let's row reduce the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -1 & -1 & 1 & 0 \\ 3 & 2 & 0 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Since c is a free variable, we have infinitely many solutions, and S is linearly dependent. This means that we can remove one of the matrices in S by the Plus/Minus theorem.

The next theorem is incredibly useful.

Theorem. Columns of a matrix have the same linear relationship as the columns of its reduced row echelon form.

From the example above, where we row reduced

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -1 & -1 & 1 & 0 \\ 3 & 2 & 0 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

We can easily see that

$$\mathbf{w}_3 = 2\mathbf{w}_1 - 3\mathbf{w}_2.$$

From the theorem, we have that

$$\mathbf{v}_3 = 2\mathbf{v}_1 - 3\mathbf{v}_2.$$

Let's verify this.

$$2\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} - 3\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = v_3.$$

Therefore, the set W is two dimensional, and the span is given by $\mathbf{v}_1$ and $\mathbf{v}_2$. Let's look at another example.

Example 24. Given the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_4 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ -1 \end{pmatrix},$$

and the vector

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Suppose that

$$\mathbf{u} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 + d\mathbf{v}_4.$$

Determine the value of d. We can begin by constructing the augmented matrix and row reducing it.

$$A = \left(\mathbf{v}_1 \mid \mathbf{v}_2 \mid \mathbf{v}_3 \mid \mathbf{v}_4 \mid \mathbf{u} \right) \quad \rightsquigarrow \quad R = \begin{pmatrix} 1 & 0 & 0 & 0 & 1/4 \\ 0 & 1 & 0 & 0 & 1/4 \\ 0 & 0 & 1 & 0 & 3/4 \\ 0 & 0 & 0 & 1 & -1/4 \end{pmatrix}.$$

We can deduce that $\mathbf{v}_1, \ldots, \mathbf{v}_4$ are linearly independent. From R, we have that

$$\mathbf{r}_5 = \cdots + -\frac{1}{4}\mathbf{r}_4,$$

and thus it follows that

$$\mathbf{u} = \cdots + -\frac{1}{4}\mathbf{v}_4.$$

Therefore, the value of the coefficient $d = -1/4$.

1.4 Subspaces of a Matrix

In this section, we will examine *column spaces* and *row spaces*.

Definition (Column Space). The *column space* of a matrix A is the space spanned by all columns of A.

Let's illustrate this with the following example.

Example 25. Let A be

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 3 & 2 \end{pmatrix}.$$

The column space of A is

$$\mathrm{Col}(A) = \mathrm{Span}\left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\} \leq \mathbb{R}^2.$$

As the final vector can be written as a linear combination of the first two, and the first two are linearly independent, the basis for the column space of A is

$$\left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right\}.$$

The dimension of the column space is the number of items in its basis

$$\dim(\mathrm{Col}(A)) = 2,$$

which is the same as its parent space, $\mathbb{R}^2$.

Let's do another simple example.

Example 26. Suppose that we are given

$$B = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since the latter two elements are linear combinations of the first, we have that

$$\mathrm{Col}(B) = \mathrm{Span}\left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\}, \quad \dim(\mathrm{Col}(B)) = 1.$$

Definition (Rank). The *rank* of A is the dimension of its column space. Equivalently, $\mathrm{rank}(A)$ is the number of linear ones in the row reduced echelon form of A.

Example 27. Find a basis for $\mathrm{Col(A)} \leq \mathbb{R}^4$ where

$$A = \begin{pmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{pmatrix}.$$

Row reducing this matrix, we obtain

$$R = \begin{pmatrix} 1 & -3 & 0 & -14 & 0 & -37 \\ 0 & 0 & 1 & 3 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

We see that the second column is redundant, as it is a linear combination of the first. The fourth column can be removed as well, as it is a linear combination of the first and third columns. The final column can also be removed, for similar reasons. Therefore, the basis for $\mathrm{Col}(A)$ is

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 4 \\ 9 \\ 9 \\ -4 \end{pmatrix}, \begin{pmatrix} 5 \\ 8 \\ 9 \\ -5 \end{pmatrix} \right\}.$$

In other terms, $\mathrm{rank}(A) = \dim(\mathrm{Col}(B)) = 3$.

Next, we introduce the counterpart to column space: the row space.

Definition (Row Space). The *row space* of a matrix A is the space spanned by all rows of A, denoted $\mathrm{row}(A)$.

Example 28. Let A be

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Find $\text{row}(A) \leq \mathbb{R}^2$ and its basis. We have that

$$\begin{aligned}\text{row}(A) &= \text{Span}\left\{\begin{pmatrix}1\\-1\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}\right\}\\ &= \text{Span}\left\{\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}\right\}\\ &= \mathbb{R}^2.\end{aligned}$$

We should also note how transposition of a matrix affects row/column space.

Remark 2. The row space of a matrix A is equivalent to the column space of A^T. In other terms,

$$\text{row}(A) = \text{col}(A^T).$$

Although we can use this method to find the row space, it is also possible to use elementary row operations, as highlighted in the lemma below.

Lemma 9. Elementary row operations do not change row space, i.e., if we have the matrix A and we apply the EROs ρ to it to obtain B

$$A \underbrace{\rightsquigarrow}_{\rho} B,$$

then $\text{row}(A) = \text{row}(B)$. In particular, if R is the row reduced echelon form of A, then

$$\text{row}(A) = \text{row}(R).$$

Example 29. Find a basis for $\text{row}(A)$, where

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{pmatrix}.$$

We have that $\text{row}(A) \leq \mathbb{R}^5$. To find the basis, we can either take A^T and find its column space, or we can simply apply elementary row operations to reduce it. Performing the latter method, we obtain

$$R = \begin{pmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

We see that the final row is redundant, and that the remaining three rows are linearly independent, and as such form a basis for $\text{row}(A)$. In conclusion

$$\begin{aligned}\text{row}(A) &= \text{row}(R)\\ &= \text{Span}\left\{\begin{pmatrix}1\\0\\0\\-2\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\\-1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\\1\\0\end{pmatrix}\right\}.\end{aligned}$$

Therefore, $\dim(\text{row}(A)) = 3$.

In the previous example, it is actually the case that $\text{rank}(A^T) = \text{rank}(A) = 3$. This is actually true in general.

Theorem. For any matrix A, we have that

$$\text{rank}(A) = \text{rank}(A^T).$$

Proof. Beginning with the right hand side, we have that

$$\begin{aligned}
\text{rank}(A^T) &= \dim(\text{col}(A^T)) \\
&= \dim(\text{row}(A)) \\
&= \dim(\text{row}(R)) && \text{(RREF)} \\
&= \#\text{ of leading 1's in RREF} \\
&= \dim(\text{col}(R)) \\
&= \text{rank}(R) \\
&= \text{rank}(A).
\end{aligned}$$

□

Another space is the *null space*, which is defined as follows.

Definition. The *null space* is the space of solutions

$$\{\mathbf{x} \mid A\mathbf{x} = \mathbf{0}\}.$$

The dimension of the null space of A is denoted $\text{nullity}(A)$.

Example 30. Given the matrix

$$A = \begin{pmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 5 \\ -1 & -3 & 3 & 0 \end{pmatrix}.$$

The null space of A, denoted $\text{null}(A)$, is

$$\text{null}(A) = \{\mathbf{x} \mid A\mathbf{x} = \mathbf{0}\} \leq \mathbb{R}^4.$$

Constructing the augmented matrix and row reducing, we have

$$\left(\begin{array}{c|c} A & \mathbf{0} \end{array} \right) \rightsquigarrow \left(\begin{array}{cccc|c} 1 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

We have the system of equations

$$\begin{aligned}
x_1 &= -3x_2 - x_4 \\
x_2 &= \text{free} \\
x_3 &= -\frac{1}{3}x_4 \\
x_4 &= \text{free}.
\end{aligned}$$

Rewriting the system in terms of free variables, we have

$$\begin{pmatrix} -3x_2 - x_4 \\ x_2 \\ -1/3x_4 \\ x_4 \end{pmatrix} = x_2 \underbrace{\begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{\mathbf{v}_1} + x_4 \underbrace{\begin{pmatrix} -1 \\ 0 \\ -1/3 \\ 1 \end{pmatrix}}_{\mathbf{v}_2}.$$

These two vectors are linearly independent and span A, therefore, the basis is

$$\{\mathbf{v}_1, \mathbf{v}_2\}.$$

The dimension of the null space is

$$\dim(\text{null}(A)) = 2.$$

Next we will introduce a very important theorem.

Theorem (Rank-Nullity Theorem). For any matrix A

$$\text{rank}(A) + \text{nullity}(A) = \#\text{ cols of } A.$$

This theorem is very powerful, as we generally always know the number of columns of A. Therefore, if we have rank, we can calculate nullity, and vice versa.

Example 31. Find the rank and nullity of A and verify the Rank-Nullity theorem in this case.

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{pmatrix}$$

Using Sage, we can calculate the reduced row echelon form

$$R = \begin{pmatrix} 1 & 0 & 0 & -2 & 3 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0. \end{pmatrix}$$

Counting the number of leading ones, we see that $\text{rank}(A) = 3$. A has five columns. Therefore, the nullity of A is 2.

Let's do another easy example.

Example 32. If A is a 7×12 matrix, what is the *largest possible dimension* of $\text{row}(A)$. Since each row has twelve entries, $\text{row}(A) \leq \mathbb{R}^{12}$. Therefore, $\dim(\text{row}(A)) \leq 12$.

One more example which shows a very practical example of the Rank-Nullity Theorem.

Example 33. Let A be an 8×6 matrix such that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. What is the rank of A? By definition, we have that $\{A\mathbf{x} = \mathbf{0}\} = \{\mathbf{0}\}$. By definition, $\dim(\text{null}(A)) =$

nullity$(A) = 0$. Therefore, by the Rank-Nullity Theorem, rank$(A) = 6$.

Let's do a past-quiz question.

Example 34. Let A be a 20×23 matrix. Given that rank$(A) = 4$, determine the nullity. By the Rank-Nullity theorem, we have that

$$\begin{aligned} \text{rank}(A) + \text{nullity}(A) &= \#\text{ columns of } A \\ 4 + \text{nullity}(A) &= 23 \\ \text{nullity}(A) &= 19. \end{aligned}$$

We will next do a past exam question.

Example 35. Let $A \in M_{m\times n}(\mathbb{R})$. Suppose that col$(A)$ is a subspace of $\mathbb{R}^9$, row(A) is a subspace of $\mathbb{R}^8$, and the nullity of A^T is 4. Determine rank(A^T) and nullity(A).

Since col$(A) \leq \mathbb{R}^9$ and row$(A) \leq \mathbb{R}^8$, we have that A is a 9×8 matrix. Let's start with rank(A^T). We can still apply the Rank-Nullity theorem here

$$\begin{aligned} \text{rank}(A^T) + \text{nullity} &= \#\text{ columns of } A \\ \text{rank}(A^T) + 4 &= 9 \\ \text{rank}(A^T) &= 5. \end{aligned}$$

Since rank$(A^T) =$ rank(A), we have that

$$\begin{aligned} \text{rank}(A) + \text{nullity}(A) &= \#\text{ cols of } A \\ 5 + \text{nullity}(A) &= 8 \\ \text{nullity}(A) &= 3. \end{aligned}$$

Chapter 2

Linear Transformations

2.1 Linear Transformations

Definition (Linear Transformation). Let V, W be vector spaces. A function $T : V \to W$ is called a *linear transformation*. If it satisfies the following two axioms:

1) $\forall \mathbf{x}, \mathbf{y} \in V$, we have that $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$.
2) $\forall \mathbf{x} \in V$, $\forall k \in \mathbb{R}$, we have that $T(k\mathbf{v}) = kT(\mathbf{v})$.

Note that in the definition above, V and W are not necessarily the same vector space, so addition/scalar multiplication between their elements may not be standard (e.g., suppose $V = \mathbb{R}^n$ and $W = \mathbf{P}_2(x)$).

Remark 3. If $T : V \to W$ is a linear transformation, then

$$T(k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n) = k_1T(\mathbf{v}_1) + \cdots + k_nT(\mathbf{v}_n).$$

That is to say, it preserves linear combinations.

Let's look at a couple of easy linear transformations.

Example 36. The identity operator $\mathrm{id} : V \to V$ simply maps a vector to itself, i.e., $\mathrm{id}(\mathbf{x}) = \mathbf{x}$.

Example 37. The zero operator $0 : V \to V$ maps any vector to the zero vector, i.e., $0(\mathbf{x}) = \mathbf{0}$.

Example 38. Let $k \in \mathbb{R}$. Define $T_k : V \to V$ to be $T_k(\mathbf{x}) = k\mathbf{x}$. This function multiplies any vector in V by a scalar k. It is called the scalar operator.

Let us consider some more non-trivial examples.

Example 39. Define the differential operator $D : \mathbf{P}_n(x) \to \mathbf{P}_n(x)$ to be $D(\mathbf{p}) = \mathbf{p}'$ (the derivative of $\mathbf{p}$).

Example 40. Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ to be

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^2 \\ y \end{pmatrix}.$$

This is not linear. Consider

$$T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix}\right) = T\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

On the other hand, we have that

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} + T\begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

Since it does not satisfy the first axiom of a linear transformation, it is not a linear transformation.

Let's move on to an problem from the example sheet.

Example 41. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a map defined below. Determine if T is a linear transformation.

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ x-2y \\ 3x \end{pmatrix}.$$

Observe that we can rewrite

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ 3 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

In a more general form, we have that $T(\mathbf{x}) = A\mathbf{x}$. This is known as a *matrix transformation*, which is always linear. Let's verify that it is a linear combination. First, we want to show $T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$.

$$\begin{aligned} T(\mathbf{x}+\mathbf{y}) &= A(\mathbf{x}+\mathbf{y}) \\ &= A\mathbf{x} + A\mathbf{y} \\ &= T(\mathbf{x}) + T(\mathbf{y}). \end{aligned}$$

The second axiom is also preserved. We want to show that $T(k\mathbf{x}) = kT(\mathbf{x})$.

$$\begin{aligned} T(k\mathbf{x}) &= A(k\mathbf{x}) \\ &= kA\mathbf{x} \\ &= kT(\mathbf{x}). \end{aligned}$$

Therefore, we can conclude that this transformation is linear.

Let's generalize the result above with a lemma.

Lemma 10. The map $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{x}) = A\mathbf{x}$ (where A is a suitable matrix of compatible dimension) is a linear transformation.

This is one of the most important linear transformations that we will study in detail in the upcoming topic. Another important property of linear transformations is the following.

Lemma 11. Let $T : V \to W$ be linear.

1) $T(\mathbf{0}_V) = \mathbf{0}_W$.
2) $T(-\mathbf{v}) = -T(\mathbf{v})$, $\forall \mathbf{v} \in V$.

Proof of 1). We know that $\mathbf{0}_V = 0 \cdot \mathbf{v}$ by properties of vector spaces. Therefore, we have

$$T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}_W.$$

□

Proof of 2). LAETR. □

Showing the two properties above is an easy test to see whether or not a function is a linear transformation. Consider the following example.

Example 42. Define $T : \mathbb{R}^2 \to \mathbb{R}^2$.

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 5 \end{pmatrix}.$$

This is not linear since

$$T\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Note that however, a mapping like

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 0 \end{pmatrix}$$

would be linear as we can rewrite it as

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

Note that when we say something is a *linear transformation*, this is not the same as saying it is a *linear function*. Linearity in Calculus and Linear Algebra are not always the same.

Remark 4. If $f : \mathbb{R} \to \mathbb{R}$ is a linear function, it is a function of the form $f(x) = ax + b$. However, in a vector space, if $b \neq 0$, then $T : \mathbb{R} \to \mathbb{R}$ defined as $T(x) = ax + b$ is *not* a linear transformation (and hence, not linear).

Example 43. Let $T : V \to \mathbb{R}$ be a linear transformation. Suppose we are given that $T(\mathbf{v}) = 2$ and $T(\mathbf{w}) = -1$. Determine $T(3\mathbf{v} - 5\mathbf{w})$.

Since T is a linear transformation, it preserves linear combinations. Therefore, we can write

$$\begin{aligned} T(3\mathbf{v} - 5\mathbf{w}) &= T(3\mathbf{v}) - T(5\mathbf{w}) \\ &= 3T(\mathbf{v}) - 5T(\mathbf{w}) \\ &= 3 \cdot (2) - 5 \cdot (-1) \\ &= 6 + 5 \\ &= 11. \end{aligned}$$

The example above shows that we don't need to know the exact linear transformation to still perform computations using it. In fact, suppose $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for V and we know $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)$. Then, we can determine $T : V \to W$. That is because for $\mathbf{v} \in V$, we have that

$$\mathbf{v} = k_1\mathbf{v}_1 + \cdots + k_n\mathbf{v}_n.$$

Therefore,

$$T(\mathbf{v}) = k_1 T(\mathbf{v}_1) + \cdots + k_n T(\mathbf{v}_n).$$

Example 44. Suppose $T : \mathbb{R}^3 \to \mathbb{R}^2$ is a linear transformation such that

$$T\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}3\\1\end{pmatrix}, \quad T\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}-1\\2\end{pmatrix}, \quad T\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}5\\0\end{pmatrix}.$$

Can we compute

$$T\begin{pmatrix}3\\-2\\1\end{pmatrix}?$$

Yes we can, we can simply decompose T as a linear combination of its basis vectors. We have that

$$\begin{aligned}
T\begin{pmatrix}3\\-2\\1\end{pmatrix} &= T\left(3\begin{pmatrix}1\\0\\0\end{pmatrix} - 2\begin{pmatrix}0\\1\\0\end{pmatrix} + 1\begin{pmatrix}0\\0\\1\end{pmatrix}\right) \\
&= 3T\begin{pmatrix}1\\0\\0\end{pmatrix} - 2T\begin{pmatrix}0\\1\\0\end{pmatrix} + T\begin{pmatrix}0\\0\\1\end{pmatrix} \\
&= 3\begin{pmatrix}3\\1\end{pmatrix} - 2\begin{pmatrix}-1\\2\end{pmatrix} + \begin{pmatrix}5\\0\end{pmatrix} \\
&= \begin{pmatrix}16\\-1\end{pmatrix}
\end{aligned}$$

Can we find $T(\mathbf{x})$ where $\mathbf{x}$ is a generic vector $(x, y, z)^T$? Yeah. Again, we can decompose $\mathbf{x}$ into its basis vectors again

$$\begin{aligned}
T\begin{pmatrix}x\\y\\z\end{pmatrix} &= T\left(x\begin{pmatrix}1\\0\\0\end{pmatrix} + y\begin{pmatrix}0\\1\\0\end{pmatrix} + z\begin{pmatrix}0\\0\\1\end{pmatrix}\right) \\
&= x\begin{pmatrix}3\\1\end{pmatrix} + y\begin{pmatrix}-1\\2\end{pmatrix} + z\begin{pmatrix}5\\0\end{pmatrix} \\
&= \begin{pmatrix}3 & -1 & 5\\1 & 2 & 0\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}.
\end{aligned}$$

Let's do some more examples.

Example 45. Is the transpose operator linear? Consider $f : M_{m\times n} \to M_{n\times m}$ where $f(A) = A^T$.

We have that

$$\begin{aligned} f(A+B) &= (A+B)^T \\ &= A^T + B^T \\ &= f(A) + f(B). \end{aligned}$$

Furthermore,

$$f(kA) = (kA)^T = kA^T = kf(A).$$

Therefore, the transpose operation is linear.

Exercise

Is the trace operation, i.e., $T(A) = \text{tr}(A)$, linear?

Example 46. Is the determinant operation, i.e., $T : M_{n\times n} \to \mathbb{R}$ where $T(A) = \det(A)$, linear? Consider

$$\begin{aligned} T\left(2\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\right) &= T\begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix} \\ &= 16 - 24 \\ &= -8. \end{aligned}$$

On the other hand, if we pull out the constant

$$\begin{aligned} 2T\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} &= 2(4-6) \\ &= -4. \end{aligned}$$

Therefore, the determinant operation is *not* linear.

Example 47. Does there exist a linear transformation $T : \mathbb{R} \to \mathbb{R}$ satisfying the following?

$$T(3) = 10, \quad T(2) = \frac{20}{3}.$$

Consider $T(3) = 10$, which is equivalent to

$$\begin{aligned} T(3\cdot 1) &= 10 \\ 3T(1) &= 10 \\ T(1) &= \frac{10}{3}. \end{aligned}$$

Since 1 is basically the 'basis vector' in the instance of $\mathbb{R}$, we can now generalize the transformation of any real number $\mathbf{x}$ to

$$T(x) = T(x\cdot 1) = xT(1) = x\cdot\frac{10}{3}.$$

Therefore, T is a linear transformation.

Lemma 12. If $B = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for V, then any linear transformation $T : V \to W$ is *uniquely* determined by $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)$.

Example 48. Let $T : \mathbb{R} \to \mathbb{R}$ be a linear transformation such that $T(5) = 3$. Find $T(7)$. We have that $T(5) = T(5 \cdot 1) = 5T(1)$, therefore, $T(1) = \frac{3}{5}$. Therefore,

$$T(7) = 7T(1) = \frac{21}{5}.$$

Let's do a past quiz question.

Example 49. Is the following a linear transformation?

$$T : M_{m\times n}(\mathbb{R}) \to \mathbb{R}; \quad T(A) = \operatorname{rank}(A).$$

This is not linear. Let's construct a counterexample. Consider I_2.

$$\begin{aligned} T\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) &= T\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\ &= 2. \end{aligned}$$

However, we have that

$$T\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + T\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 2 + 2 = 4,$$

which is impossible, as we cannot have rank greater than $\min(n, m)$ (which in this case, is 2).

Example 50. Let $T : \mathbf{P}_2(\mathbf{x}) \to \mathbb{R}$ be a linear transformation such that

$$T(2x^2) = 6, \quad T(3x) = 12, \quad T(4) = 20.$$

Determine $T(4x^2 + 2x + 3)$. From $T(2x^2) = 6$, we have that $T(x^2) = 3$. From $T(3x) = 12$, we have that $T(x) = 4$. Finally, $T(4) = 20$ implies that $T(1) = 5$. Therefore

$$\begin{aligned} T(4x^2 + 2x + 3) &= T(4x^2) + T(2x) + T(3) \\ &= 4T(x^2) + 2T(x) + 3T(1) \\ &= 4 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 \\ &= 12 + 8 + 15 \\ &= 35. \end{aligned}$$

2.2 Matrix Transformations

Recall that a transformation $T : \mathbb{R}^m \to \mathbb{R}^m$ of the form $\mathbf{x} \to A\mathbf{x}$ is called a *matrix transformation*. We also showed in a lemma that every matrix transformation is linear. Our goal for today is to show the converse of that statement, i.e., if $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then T is a matrix transformation.

Example 51. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation such that

$$T\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}, \quad T\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}, \quad T\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix}.$$

We will show that T is a matrix transformation. Let $\mathbf{x} \in \mathbb{R}^3$, i.e.,

$$\mathbf{x} = \begin{pmatrix}x\\y\\z\end{pmatrix}.$$

We can decompose $\mathbf{x}$ as follows

$$\mathbf{x} = x\begin{pmatrix}1\\0\\0\end{pmatrix} + y\begin{pmatrix}0\\1\\0\end{pmatrix} + z\begin{pmatrix}0\\0\\1\end{pmatrix}.$$

Applying the transformation T on both sides, we obtain

$$T(\mathbf{x}) = x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}1\\1\end{pmatrix} + z\begin{pmatrix}0\\-1\end{pmatrix}.$$

This is equivalent to

$$\begin{aligned} T(\mathbf{x}) &= \begin{pmatrix}x+y\\y-z\end{pmatrix} \\ &= \begin{pmatrix}1 & 1 & 0\\0 & 1 & -1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}. \end{aligned}$$

Since $T(\mathbf{x})$ has output of the form $A\mathbf{x}$, it is indeed a matrix transformation.

2.2.1 Standard Matrix

Let's formalize this concept.

Theorem. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then there exists a unique matrix A such that $T(\mathbf{x}) = A\mathbf{x}$. The matrix A is called the *standard matrix* for T.

Proof. Let $\mathbf{x} \in \mathbb{R}^n$. Recall that $\mathbb{R}^n$ has the standard basis $\mathbf{e}_j$, the column vector which is all zeros

except for a 1 in the jth row. We have that

$$\begin{aligned}
\mathbf{x} &= \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\
\mathbf{x} &= x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n \\
T(\mathbf{x}) &= x_1 T(\mathbf{e}_1) + \cdots + x_n T(\mathbf{e}_n) \\
T(\mathbf{x}) &= \left(\begin{array}{c|c|c|c} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{array} \right) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\
T(\mathbf{x}) &= A\mathbf{x}.
\end{aligned}$$

□

Let's use the method applied in the theorem above to find the standard matrix of a linear transformation.

Example 52. Define a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

We will determine its standard matrix. From the theorem above, there exists A such that $T(\mathbf{x}) = A\mathbf{x}$. We have that

$$\begin{aligned}
A &= \left(\begin{array}{c|c|c} T(\mathbf{e}_1) & T(\mathbf{e}_2) & T(\mathbf{e}_3) \end{array} \right) \\
&= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.
\end{aligned}$$

Let's double check.

$$\begin{aligned}
T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\
&= \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.
\end{aligned}$$

Let's do one more example.

Example 53. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation such that

$$T\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}.$$

We will find a standard matrix for T. We know that

$$\begin{aligned} A &= \left(T\begin{pmatrix}1\\0\end{pmatrix} \,\middle|\, T\begin{pmatrix}0\\1\end{pmatrix} \right) \\ &= \begin{pmatrix} ? & -1 \\ ? & 2 \\ ? & 1 \end{pmatrix} \end{aligned}$$

To find the unknown, we want to find

$$\begin{pmatrix}1\\0\end{pmatrix} = a\begin{pmatrix}1\\1\end{pmatrix} + b\begin{pmatrix}0\\1\end{pmatrix}.$$

Therefore $(a,b) = (1,-1)$. So

$$\begin{aligned} \begin{pmatrix}1\\0\end{pmatrix} &= \begin{pmatrix}1\\1\end{pmatrix} - \begin{pmatrix}0\\1\end{pmatrix} \\ T\begin{pmatrix}1\\0\end{pmatrix} &= T\begin{pmatrix}1\\1\end{pmatrix} - T\begin{pmatrix}0\\1\end{pmatrix} \\ &= \begin{pmatrix}1\\2\\3\end{pmatrix} - \begin{pmatrix}-1\\2\\1\end{pmatrix} \\ &= \begin{pmatrix}2\\0\\2\end{pmatrix}. \end{aligned}$$

Therefore, we have the standard matrix

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \\ 2 & 1 \end{pmatrix}.$$

There is actually a programmatic way to find the standard matrix.

Ajarn Pornrat's Method for Finding Standard Matrix

Let's redo example 25 using a more efficient method. We know that

$$T\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}, \quad T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-1\\2\\1\end{pmatrix}.$$

We know that $T(\mathbf{x}) = A\mathbf{x}$ for some A. Therefore, it must be the case that

$$A\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}, \quad A\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-1\\2\\1\end{pmatrix}.$$

In other terms

$$A\begin{pmatrix}1 & 0\\1 & 1\end{pmatrix} = \begin{pmatrix}1 & -1\\2 & 2\\3 & 1\end{pmatrix}.$$

Recall that the inverse of a 2×2 matrix A is

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Now we can solve for A by multiplying the inverse of the matrix on the left.

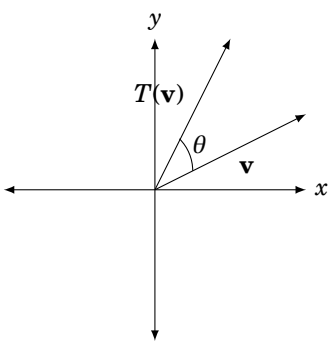
$$\begin{aligned} A &= \begin{pmatrix} 1 & -1 \\ 2 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} 1 & -1 \\ 2 & 2 \\ 3 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 \\ 2 & 2 \\ 3 & 1 \end{pmatrix} \cdot \frac{1}{1} \cdot \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -1 \\ 0 & 2 \\ 2 & 1 \end{pmatrix}. \end{aligned}$$

This method is nice in the sense that it is can be done programmatically.

2.2.2 Examples of Linear Transformations in $\mathbb{R}^2$

Rotation Transformation

One of the most important transformations is a *rotation matrix* $T_\theta : \mathbb{R}^2 \to \mathbb{R}^2$, which rotates a vector $\mathbf{x} \in \mathbb{R}^2$ by θ radians counter-clockwise.



The standard matrix for this transformation is

$$A_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

which is sometimes called the *rotation matrix*.

Example 54. If we wanted to rotate $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ by $\frac{\pi}{3}$, we could do so by multiplying the matrix

$$\begin{aligned} A_{\pi/3} &= \begin{pmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3} \\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}. \end{aligned}$$

So we'd have

$$\begin{aligned} A\mathbf{v} &= \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & -\sqrt{3} \\ \frac{3}{\sqrt{2}} & 1 \end{pmatrix}. \end{aligned}$$

Reflection Transformation

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ reflect a vector $\mathbf{x} \in \mathbb{R}^2$ across the x-axis, i.e.,

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Note how we can easily determine the standard matrix in this instance. A standard matrix for reflection about the y-axis can be found similarly.

Projection

Another important transformation is the projection of a vector. The projection of onto the y-axis is defined as the following transformation

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ y \end{pmatrix}.$$

The standard matrix for this projection is

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

as we have that

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Example 55. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a reflection about the line $y = 2x$. Determine the standard matrix for T.

Let's construct two "nice" vectors, i.e. vectors which we can easily compute the image for, that will allow us to determine the transformation. For example, consider the vector on the line

$$\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

We can also find a vector perpendicular to this

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

We have that

$$T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad T\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

We know that

$$A\mathbf{v} = \mathbf{v}, \quad A\mathbf{u} = -\mathbf{u}.$$

So we can rewrite

$$\begin{aligned} A\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} &= \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \\ A &= \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}^{-1} \\ A &= \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \cdot -\frac{1}{5} \cdot \begin{pmatrix} -1 & -2 \\ -2 & 1 \end{pmatrix} \\ A &= -\frac{1}{5}\begin{pmatrix} 3 & -4 \\ -4 & -3 \end{pmatrix} \end{aligned}$$

2.2.3 Composing Linear Transformations

Suppose that $T : \mathbb{R}^k \to \mathbb{R}^n$ and $S : \mathbb{R}^n \to \mathbb{R}^m$. The *composition* of S and T, denoted $S \circ T$, is a transformation from $\mathbb{R}^k \to \mathbb{R}^m$. However, if T and S have standard matrices A and B, respectively, what is the standard matrix of their composition? We have that

$$\begin{aligned} (S \circ T)(\mathbf{x}) &= S(T(\mathbf{x})) \\ &= S(A\mathbf{x}) \\ &= B(A\mathbf{x}) \\ &= (BA)\mathbf{x}. \end{aligned}$$

In this way, we can work out transformations for complicated actions, e.g., a reflection and then a rotation.

Example 56. Given the vector

$$\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

rotate by 90 degrees counter-clockwise and then reflect the resulting vector about the x-axis. Suppose the final vector is $\mathbf{w}$. Determine the vector $\mathbf{w}$ and find a matrix C such that $C\mathbf{v} = \mathbf{w}$.

Recall that the standard matrix for a rotation is

$$A_{90} = \begin{pmatrix} \cos 90 & -\sin 90 \\ \sin 90 & \cos 90 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

A reflection about the x-axis has the standard matrix

$$B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore, we have that

$$\begin{aligned}\mathbf{w} &= BA\mathbf{v}\\ &= \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1\\ 2\end{pmatrix}\\ &= \begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\begin{pmatrix}1\\ 2\end{pmatrix}\\ &= \begin{pmatrix}-2\\ -1\end{pmatrix}\end{aligned}$$

2.3 Kernel and Image

2.3.1 Terminology

Let's do a quick recap of some function related terminology.

Definition (Image/Preimage). Suppose we have a function $f : X \to Y$. Suppose $f(a) = b$. We say that b is the *image* of a under f. On the other hand, a is a *preimage* of b (under f).

We also have the concept of images of a function.

Definition (Image/Preimage of a Function). Let $f : X \to Y$. The *image* of f is the set

$$f(X) = \{f(x) \mid x \in X\}.$$

In other words, it is the set of all possible images in the function. Let $B \subseteq Y$. The *preimage* of B is the set

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}.$$

In particular, if B is a singleton, i.e., $B = \{b\}$, then

$$\begin{aligned}f^{-1}(b) &= f^{-1}(\{b\})\\ &= \{x \in X \mid f(x) = b\}.\end{aligned}$$

Let's do a quick example.

Example 57. Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f = x^2$. The image of 5 is 25, as $f(5) = 25$. The image of $\mathbb{R}$ is $[0, \infty)$. The preimage of 9 is $f^{-1}(9) = \{-3, 3\}$. The preimage of $\{36, 100\}$ is

$$\begin{aligned}f^{-1}(\{36, 100\}) &= \{x \in \mathbb{R} \mid x^2 \in \{36, 100\}\}\\ &= \{\pm 6, \pm 10\}.\end{aligned}$$

Example 58. Let $D : \mathbf{P}_3(x) \to \mathbf{P}_3(x)$ be defined as the differential operation $D(p) = p'$. Sup-

pose that $p(x) = 6x^2 + 5$. The preimage of p is

$$\begin{aligned} D^{-1}(p) &= \{f \in \mathbf{P}_3(x) \mid f'(x) = 6x^2 + 5\} \\ &= \left\{\int (6x^2 + 5)\,dx\right\} \\ &= \{2x^3 + 5x + C \mid c \in \mathbb{R}\} \end{aligned}$$

This terminology also applies to linear transformations.

2.3.2 Kernel and Image

Now that we have refreshed our understanding of terminology, let's move on to the main topic for today. The first is the *kernel* of a linear transformation, which is the preimage of the zero vector.

Definition (Kernel). Let $T : V \to W$ be a linear transformation. We define the *kernel* of T to be

$$T^{-1}\{\mathbf{0}\} = \ker(T) = \{v \in V : T(\mathbf{v}) = \mathbf{0}_W\}.$$

Note that $\mathbf{0}_V \in \ker(T)$.

We also define the *image* of a linear transformation, which is similar to that of a function's.

Definition (Image). Let $T : V \to W$ be a linear transformation. The *image* of T is

$$\mathrm{Im}(T) = \{T(\mathbf{v}) : v \in V\}.$$

Example 59. Let $D : \mathbf{P}_3(x) \to \mathbf{P}_3(x)$ be the differential operator. We have that

$$\ker(D) = \{p \in \mathbf{P}_3(x) \mid p' = 0\} = \{c \mid c \in \mathbb{R}\}.$$

So the kernel is the set of all constants. As for the image, we have that

$$\begin{aligned} \mathrm{Im}(D) &= \{(ax^3 + bx^2 + cx + d)' \mid a, b, c, d \in \mathbb{R}\} \\ &= \{3ax^2 + 2bx + c \mid a, b, c \in \mathbb{R}\} \\ &= \{\alpha x^2 + \beta x + \gamma \mid \alpha, \beta, \gamma \in \mathbb{R}\}, \end{aligned}$$

so the image is the set of all degree two polynomials. Observe that the kernel is in the domain but the image is in the codomain.

We now note the important result that we observed in the previous example.

Theorem. Let $T : V \to W$ be linear.

1) $\ker(T) \leq V$
2) $\mathrm{Im}(T) \leq W$

Proof.

1) We can use the subspace test. Obviously, $T(\mathbf{0}) = \mathbf{0}$ so $\mathbf{0} \in \ker(T)$. If $\mathbf{u}, \mathbf{v} \in \ker(T)$, we have that since T is linear

$$\begin{aligned} T(\mathbf{u}+\mathbf{v}) &= T(\mathbf{u}) + T(\mathbf{v}) \\ &= \mathbf{0} + \mathbf{0} \\ &= \mathbf{0}. \end{aligned}$$

Therefore, $\mathbf{u}+\mathbf{v} \in \ker(T)$. Finally, if $\mathbf{u} \in \ker(T)$ and $\lambda \in \mathbb{R}$, we have that

$$\begin{aligned} T(\lambda\mathbf{u}) &= \lambda T(\mathbf{u}) \\ &= \lambda\mathbf{0} \\ &= \mathbf{0}. \end{aligned}$$

Therefore $\lambda\mathbf{u} \in \ker(T)$. Hence, $\ker(T) \leq V$.

2) Left as an exercise to the reader.

□

Since the kernel and image of a linear transformation are subspaces, they are vector spaces. This implies they also have dimensions, nullity, etc.

Definition. Let $T : V \to W$ be a linear transformation.

- The *nullity* of T is $\dim(\ker(T))$.
- The *rank* of T is $\dim(\mathrm{Im}(T))$.

Example 60. Continuing with our differential operator $D : \mathbf{P}_3(x) \to \mathbf{P}_3(x)$, we have that

$$\ker(D) = \{c \mid c \in \mathbb{R}\} = \mathrm{span}\{1\}.$$

Therefore, $\mathrm{nullity}(D) = 1$. As for the rank, recall that

$$\mathrm{Im}(D) = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\} = \mathrm{Span}\{1, x, x^2\}.$$

Therefore, $\mathrm{rank}(D) = 3$. Observe how the rank nullity theorem is still preserved.

Before continuing, we must be careful of the fact that the preimage of elements in the codomain is not always a subspace of the domain, even if it is a subset.

Remark 5. Let $T : V \to W$ be a linear transformation. If we take $S \subset W$ and look at $T^{-1}(S) \subset V$, this may not be a subspace of V.

We can show this with a counter-example.

Example 61. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a projection onto the x-axis, i.e.,

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ 0 \end{pmatrix}.$$

Suppose we take

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

We have that the preimage of S is

$$T^{-1}(S) = \left\{ \begin{pmatrix} 1 \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\},$$

which is clearly not a subspace of $\mathbb{R}^2$ as it does not contain the zero vector.

There is also a connection to the null space and column space.

Example 62. Suppose $T : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation defined by

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

We want to find the kernel of T.

$$\begin{aligned} \ker(T) &= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} : A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\} \\ &= \operatorname{null}(A). \end{aligned}$$

The reduced row-echelon form of A is

$$R = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

so we have the system of equations

$$\begin{aligned} x_1 &= x_2 \\ x_2 &= x_2 \\ x_3 &= 0. \end{aligned}$$

Rewriting in terms of the free variable x_2, we have solutions of the form

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Therefore, we have that

$$\ker(T) = \operatorname{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

We also obtain that $\text{nullity}(T) = 1$. How about the image? We have that

$$\begin{aligned}\text{Im}(T) &= \{T(\mathbf{x}) \mid x \in \mathbb{R}^3\} \\ &= \{A\mathbf{x} \mid x \in \mathbb{R}^3\} \\ &= \text{col}(A) \\ &= \text{Span}\left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}.\end{aligned}$$

We also have that $\text{rank}(T) = 2$.

Let's formalize this.

Theorem. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation satisfying

$$T(\mathbf{x}) = A\mathbf{x}, \quad \exists! A.$$

Then, we have that

$$\begin{aligned}\ker(T) &= \text{null}(A) \\ \text{Im}(T) &= \text{col}(A).\end{aligned}$$

2.3.3 The Dimension Theorem

Next, we will introduce a very important result, which is essentially the analog of the Rank-Nullity Theorem but for linear transformations. Recall that the Rank Nullity Theorem states that for a matrix $A_{m\times n}$

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A. In this instance, we would have

- $\text{rank}(A) = \text{rank}(T)$
- $\text{nullity}(A) = \text{nullity}(T)$
- $n = \dim(\mathbb{R}^n) = \dim(\text{domain})$.

With this in mind, we introduce the dimensionality theorem.

Theorem (Dimension Theorem). Suppose $T : V \to W$ is a linear transformation and suppose $\ker(T), \text{Im}(T)$ are both finite-dimensional. Then, V is also finite dimensional and

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

In the special case where $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, we would have Rank-Nullity Theorem.

2.3.4 Injectivity and Surjectivity

Definition (Surjection, Injection, Bijection). Let $T : V \to W$ be a linear transformation.

- T is *injective* (one-to-one) if
$$T(\mathbf{u}) = T(\mathbf{v}) \implies \mathbf{u} = \mathbf{v}.$$
- T is *surjective* (onto) if
$$\forall \mathbf{w} \in W, \exists v \in V \text{ s.t. } T(\mathbf{v}) = \mathbf{w}.$$
- T is *bijective* if it is both injective and surjective.

The process for determining whether a linear transformation is surjective, injective, or bijective is much simpler than those for any regular functions.

Theorem (Criteria for Injectivity). Let $T : V \to W$ be a linear transformation. Then, T is injective if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof.

($\Leftarrow$)

Assume that $\ker(T) = \{\mathbf{0}\}$. Suppose $T(\mathbf{u}) = T(\mathbf{v})$. We want to show that $\mathbf{u} = \mathbf{v}$.

$$\begin{aligned} T(\mathbf{u}) &= T(\mathbf{v}) \\ T(\mathbf{u}) - T(\mathbf{v}) &= \mathbf{0} \\ T(\mathbf{u} - \mathbf{v}) &= \mathbf{0}. \end{aligned}$$

Since $\mathbf{u} - \mathbf{v} \in \ker(T) = \{\mathbf{0}\}$, we have that

$$\mathbf{u} - \mathbf{v} = \mathbf{0} \iff \mathbf{u} = \mathbf{v}.$$

($\Rightarrow$)

Left as an exercise to the reader. □

Theorem (Criteria for Surjectivity). Let $T : V \to W$ be a linear transformation and suppose W is finite dimensional. T is surjective if and only if $\mathrm{Im}(T) = W$. Alternatively, T is surjective if and only if $\mathrm{rank}(T) = \dim(W)$, i.e., T has maximum rank.

Example 63. Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation that maps

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \longmapsto \begin{pmatrix} x_1 - x_2 + 2x_3 \\ x_1 + x_2 - x_3 \\ 2x_1 - x_3 \\ 2x_2 - x_3 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 1 & -1 \\ 2 & 0 & -1 \\ 0 & 2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

We will determine whether this linear transformation is injective, surjective, or both. Row

reducing the standard matrix, we obtain

$$A \quad \rightsquigarrow \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

We have that

$$\operatorname{rank}(T) = \operatorname{rank}(A) = 3 < 4,$$

so we can conclude that T is not surjective. By the rank-nullity theorem, we see that as all columns in the RREF are pivot columns

$$\operatorname{nullity}(T) = \operatorname{nullity}(A) = 0,$$

meaning that $\ker(T) = \{\mathbf{0}\}$, so T is injective.

Example 64. Does there exists a linear transformation $T : \mathbb{R}^6 \to \mathbb{R}^7$ that is surjective? No. If T were surjective, then $\operatorname{rank}(T) = 7$. The dimension theorem implies that

$$\begin{aligned} \operatorname{rank}(T) + \operatorname{nullity}(T) &= 6 \\ 7 + \operatorname{nullity}(T) &= 6 \\ \operatorname{nullity}(T) &= -1, \end{aligned}$$

which is a contradiction, as the nullity cannot be negative.

Example 65. Does there exist a linear transformation $T : \mathbf{M}_{2\times 4} \to \mathbf{P}_8$. Again, no, for reasons similar to the previous example ($\dim(\mathbf{P}_8) = 9$).

Example 66. A linear transformation $T : \mathbb{P}_6(x) \to \mathbb{R}^4$ is surjective. What is the nullity of T? Since T is surjective,

$$\operatorname{rank}(T) = \dim(\mathbb{R}^4) = 4.$$

By the dimension theorem,

$$\begin{aligned} \operatorname{rank}(T) + \operatorname{nullity}(T) &= \dim(\mathbf{P}_6(x)) \\ 4 + \operatorname{nullity}(T) &= 7 \\ \operatorname{nullity}(T) &= 3. \end{aligned}$$

This implies also that T is not injective, as $\operatorname{nullity}(T) \neq 0$.

Chapter 3

Eigenvectors and Eigenvalues

3.1 Eigenvectors and Eigenvalues

Let's begin with an example/motivation. Suppose we have a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ which reflects a vector about the line $y = x$. Since this is a matrix transformation, we know that $T(\mathbf{x}) = A\mathbf{x}$, where the standard matrix A is

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

There are some vectors that are "invariant" under T – in this instance, meaning that for some vectors $\mathbf{v}$, $T(\mathbf{v}) = k\mathbf{v}$ (i.e., they are scaled but do not change direction). In the context of our current example, consider a vector on the line $y = x$

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \implies T(\mathbf{v}) = (1) \cdot \mathbf{v}.$$

We can also consider a vector perpendicular to the line $y = x$

$$\mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \implies T(\mathbf{w}) = (-1) \cdot \mathbf{w}$$

We call these "nice" vectors *eigenvectors*. More formally, we define them as follows.

Definition (Eigenvector/Eigenvalue). Let A be a $n \times n$ matrix. We say that $\mathbf{v} \neq \mathbf{0}$ is an *eigenvector* if

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \text{for some scalar } \lambda.$$

The scalar λ is called the *eigenvalue* associated to eigenvector $\mathbf{v}$.

In the example we just did above, we obtain two eigenvectors $\mathbf{v}$ and $\mathbf{w}$ with associated eigenvalues 1 and -1, respectively. Are these the only eigenvectors? No, there are infinitely many.

Remark 6. If $\mathbf{v}$ is an eigenvector with eigenvalue λ, then $k\mathbf{v}$ is also an eigenvector with the same eigenvalue λ.

Proof.

$$\begin{aligned} A(k\mathbf{v}) &= kA\mathbf{v} \\ &= k\lambda\mathbf{v} \\ &= \lambda(k\mathbf{v}). \end{aligned}$$

□

3.1.1 Computing Eigenvalues

By definition, we have that for $\mathbf{v} \neq 0$

$$\begin{aligned} A\mathbf{v} &= \lambda\mathbf{v} \\ (A - \lambda I)\mathbf{v} &= 0. \end{aligned}$$

Since $\mathbf{v} \neq 0$, we require $A - \lambda I$ to be singular (non-invertible).

Lemma 13. λ is an eigenvalue of A if and only if $A - \lambda I$ is singular (i.e., $\det(A - \lambda I) = 0$ or the rows are linearly dependent).

Let's illustrate this using our first example.

Example 67. Given the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

we will find its eigenvalues. We have that

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix}. \end{aligned}$$

We require the determinant of this to be zero, i.e.,

$$\det(A - \lambda I) = \lambda^2 - 1 = 0.$$

This equation has two solutions, $\lambda = 1, -1$. These are all possible eigenvalues of A. To find the corresponding eigenvectors, we go one eigenvalue at a time. For $\lambda = 1$, we have

$$\begin{aligned} A\mathbf{v} &= \mathbf{v} \\ (A - I)\mathbf{v} &= 0. \end{aligned}$$

The eigenvectors are any nonzero vectors in the null space of $A - I$, which is

$$A - I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Row reducing this, we obtain

$$\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

We have that $x = y$, so

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ y \end{pmatrix} = y\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Therefore,

$$\text{Null}(A - I) = \text{Span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\},$$

and the corresponding eigenvector to eigenvalue $\lambda = 1$ is

$$\begin{pmatrix}1\\1\end{pmatrix}.$$

We can find the other eigenvector using the same method.

The reason we ended up with two solutions is because we ended up solving an equation of degree two. This actually follows from the Fundamental Theorem of Algebra. Let's formalize this.

Lemma 14. If $A_{n\times n}$, then A has at most n distinct eigenvalues.

3.1.2 Characteristic Polynomial

Definition (Characteristic Polynomial). Let A be a $n \times n$ matrix. The *characteristic polynomial* of A is the polynomial

$$C_A(x) = \det(xI - A).$$

The *characteristic equation* is

$$C_A(x) = 0.$$

Note the following

1. The roots of $C_A(x)$ are the eigenvalues of A.

$$\begin{aligned} x \text{ is eval} &\iff \det(A - xI) = 0 \\ &\iff \det(xI - A) = 0. \end{aligned}$$

2. $C_A(x)$ is a polynomial of degree n.

Let's do an example.

Example 68. Find $C_A(x)$ given

$$A = \begin{pmatrix}3 & -2\\1 & 0\end{pmatrix}.$$

We have that

$$\begin{aligned} C_A(x) &= \det(xI - A) \\ &= \det\left(\begin{pmatrix}x & 0\\0 & x\end{pmatrix} - \begin{pmatrix}3 & -2\\1 & 0\end{pmatrix}\right) \\ &= \det\begin{pmatrix}x-3 & 2\\-1 & x\end{pmatrix} \\ &= (x-3)x + 2 \\ &= x^2 - 3x + 2. \end{aligned}$$

As we can factor x^2-3x+2 into $(x-2)(x-1)$, we can also conclude that A has the eigenvalues 2 and 1.

We also note the relationship between the characteristic equation of the transpose of a matrix.

Lemma 15. $C_A(x) = C_{A^T}(x)$.

One immediate result from this lemma is the following.

Corollary. A and A^T have the exactly same eigenvalues.

Proof.

$$\begin{aligned} C_{A^T}(x) &= \det\left(xI - A^T\right) \\ &= \det\left((xI - a^T)^T\right) \\ &= \det(xI - A) \\ &= C_A(x). \end{aligned}$$

□

Another extremely important result (that we will not prove) is the following.

Theorem (Cayley-Hamilton). A square matrix satisfies its own characteristic equation. In other terms, if $C_A(x)$ is a characteristic polynomial of A, then $C_A(A) = \mathbf{0}$.

Example 69. Suppose that

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$$

has the characteristic polynomial

$$C_A(x) = x^2 - 3x + 2.$$

Cayley-Hamilton tells us that $C_A(A) = \mathbf{0}$, i.e.,

$$\begin{aligned} A^2 - 3A + 2I &= \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} - 3\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} + 2I \\ &= \begin{pmatrix} 7 & -6 \\ 3 & -2 \end{pmatrix} + \begin{pmatrix} -9 & 6 \\ -3 & 0 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

As a consequence of this, we have that

$$\begin{aligned} A^2 - 3A + 2I &= 0 \\ 2I &= 3A - A^2 \\ 2I &= (3I - A)A \\ I &= \left(\frac{3}{2}I - \frac{1}{2}A\right)A. \end{aligned}$$

In other terms,

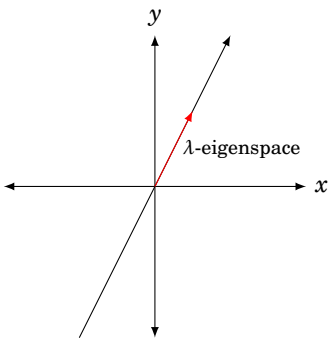
$$A^{-1} = \frac{3}{2}I - \frac{1}{2}A$$

3.1.3 Eigenspaces

Definition (Eigenspace). Suppose that λ is an eigenvalue of a matrix A. Then, the eigenspace corresponding to λ, sometimes called the λ-eigenspace, is

$$E_\lambda = \{\mathbf{v} \mid (A - \lambda I)\mathbf{v} = \mathbf{0}\} = \text{Null}(A - \lambda I).$$

Note that eigenvectors corresponding to λ are the nonzero vectors in E_λ. Geometrically, in $\mathbb{R}^2$, a one-dimensional λ-eigenspace would be a line in which applying A to a vector would only scale said vector along the line.



Example 70. Let

$$A = \begin{pmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{pmatrix}$$

with the characteristic equation

$$C_X(A) = (x-1)(x+3)^2 \implies \lambda = 1, -3.$$

For $\lambda = 1$, we have that

$$\begin{aligned} E_\lambda &= \text{Null}(A - 1I) \\ &= \text{Null}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \qquad \text{(RREF)} \end{aligned}$$

From this, we have that

$$\begin{aligned} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} &= \begin{pmatrix} -2x_3 \\ -x_3 \\ x_3 \end{pmatrix} \\ &= x_3 \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}. \end{aligned}$$

Therefore, we can conclude that

$$E_1 = \text{Span}\left\{\begin{pmatrix}-2\\-1\\1\end{pmatrix}\right\}.$$

For $\lambda = -3$, we have

$$\begin{aligned} E_{-3} &= \text{Null}(A+3I) \\ &= \text{Null}\begin{pmatrix}1 & 1 & 2\\0 & 0 & 0\\0 & 0 & 0\end{pmatrix}. \qquad \text{(RREF)} \end{aligned}$$

From this, we have

$$\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = x_2\begin{pmatrix}-1\\1\\0\end{pmatrix} + x_3\begin{pmatrix}-2\\0\\1\end{pmatrix}.$$

Therefore

$$E_{-3} = \text{Span}\left\{\begin{pmatrix}-1\\1\\0\end{pmatrix}, \begin{pmatrix}-2\\0\\1\end{pmatrix}\right\}$$

The geometrical interpretation for E_{-3} in the previous example would be than any vectors on the plane E_{-3} would be reflected and scaled by 3 on the plane if they had A applied to them. Let's cover a few more useful results.

Lemma. If A is a triangular matrix (either upper-triangular or lower-triangular), then the eigenvalues of A are the main diagonal elements.

Example 71. Given the matrix

$$A = \begin{pmatrix}5 & 1 & 2\\0 & 3 & 2\\0 & 0 & -2\end{pmatrix},$$

we can instantly determine that it has eigenvalues 5, 3, and -2.

Another useful result is the following.

Lemma 16. A square matrix A has eigenvalue 0 if and only if A is singular.

Proof.

$$\begin{aligned} \lambda = 0 &\iff \det(A - 0I) = 0 \\ &\iff \det(A) = 0 \\ &\iff A \text{ singular.} \end{aligned}$$

□

On the other hand, if we have the set of all eigenvalues of a matrix and zero is not among them, we can conclude that A is invertible.

Theorem (Trace/Determinant from Eigenvalues). Let A be a $n \times n$ matrix and let $\lambda_1, \ldots, \lambda_n$ be (not necessarily distinct) eigenvalues of A. Then

1. $\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$
2. $\det(A) = \lambda_1 \cdot \lambda_2 \ldots \lambda_n$

This theorem is useful for solving all sorts of problems, for example the below.

Example 72. Let $A_{3\times 3}$ with $\operatorname{tr}(A) = 3$. If we have two eigenvalues of A, 2 and -1, the third can be found as follows

$$\operatorname{tr}(A) = 2 - 1 + \lambda \iff \lambda = 2.$$

The determinant of the matrix would be

$$\det(A) = 2 \cdot 2 \cdot -1 = -4.$$

Let's do some past quiz questions.

Example 73. Let $A_{3\times 3}$ with $\operatorname{tr}(A) = 10$. Suppose two eigenvalues are $1, 2$. Find $\det(A)$. We know that

$$\lambda + 1 + 2 = 10 \iff \lambda = 7.$$

Therefore, the determinant is

$$\det(A) = 1 \cdot 2 \cdot 7 = 14.$$

Example 74. Let A be a 3×3 matrix with $\det(A) = 36$. Suppose we know two eigenvalues of A, 1 and 2. Find $\operatorname{tr}(A)$. Since

$$\det(A) = 1 \cdot 2 \cdot \lambda = 36,$$

we have that $\lambda = 18$. Therefore, the trace is

$$\operatorname{tr}(A) = 1 + 2 + 18 = 21.$$

Example 75. Given a 3×3 matrix A with $C_A(x) = x^3 + x^2 - 12x$. We know that the eigenvalues are the roots of this polynomial, so let's factor it.

$$\begin{aligned} x^3 + x^2 - 12x &= x(x^2 + x - 12) \\ &= x(x+4)(x-3). \end{aligned}$$

From this factorization, we can conclude that A has eigenvalues $0, -4$, and 3. From this, we can also conclude that A is not invertible.

3.2 Diagonalization

Diagonalization is a process of turning a square matrix into a *diagonal matrix*.

Definition. A $n \times n$ matrix A is *diagonalizable* if $\exists P$ invertible such that

$$P^{-1}AP = D,$$

where D is diagonal. Such a matrix P is called a *modal matrix*.

Diagonal matrices are often easier to deal with – e.g., raising them to higher powers only involves raising the elements along the diagonal to that power.

Example 76. Suppose we have a 2×2 matrix

$$A = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}.$$

We will diagonalize it. Let

$$P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

First, let's find the inverse of P.

$$P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

Then, we have that

$$\begin{aligned} P^{-1}AP &= \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 5 & 4 \\ 5 & 8 \end{pmatrix} \\ &= \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}. \end{aligned}$$

Therefore, we see that A is diagonalizable and P is a modal matrix.

We should be able to find the modal matrix ourselves, as it will not always be given.

3.2.1 Finding a Modal Matrix

Observe that we can rewrite the equation above as the following

$$P^{-1}AP = D \iff AP = PD.$$

We regard P as a matrix of columns, i.e.,

$$P = \left(\begin{array}{c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{array} \right).$$

Multiplying A to this, we would obtain

$$AP = \left(\begin{array}{c|c|c|c} A\mathbf{v}_1 & A\mathbf{v}_2 & \dots & A\mathbf{v}_n \end{array} \right).$$

Let's consider the right hand side. For D, we have

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda_n \end{pmatrix}$$

$$PD = \left(\begin{array}{c|c|c|c} \lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \dots & \lambda_n\mathbf{v}_n \end{array} \right).$$

Therefore, in order for $AD = PD$ to be satisfied, we require

$$\mathbf{A}\mathbf{v}_1 = \lambda_1 \mathbf{v}_1, \quad \ldots, \quad A\mathbf{v}_n = \lambda_n \mathbf{v}_n.$$

We can take $\mathbf{v}_i$'s as eigenvectors and λ_i's as eigenvalues. We would like P to be invertible*, so, we would not like $\mathbf{v}_i$'s to be zero vectors. Thus, they must be eigenvectors. Let's formalize this statement.

Theorem. $A_{n\times n}$ is diagonalizable if and only if A has n linearly independent eigenvectors. In the case that A is diagonalizable

$$P = \left(\begin{array}{c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_n \end{array} \right), \quad D = \begin{pmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda_n \end{pmatrix}$$

where $\mathbf{v}_i$'s are eigenvectors and λ_i's are corresponding eigenvalues.

Let's do an example to apply this.

Example 77. For the matrix

$$A = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix},$$

find P and D such that $P^{-1}AP = D$. A has the characteristic equation

$$C_A(x) = x^2 - 9x + 20 = (x-4)(x-5),$$

so A has the eigenvalues 4 and 5. Let's determine the corresponding eigenvectors. For $\lambda = 5$, the RREF for $A - 5I$ is

$$\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

Since we have that $x = y$

$$E_5 = \text{Span}\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

Similarly for $\lambda = 4$, the RREF of $A - 4I$ is

$$\begin{pmatrix} 1 & 1/2 \\ 0 & 0 \end{pmatrix}.$$

Therefore,

$$E_4 = \text{Span}\left\{ \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} \right\} = \text{Span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}.$$

Therefore, we have the eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

corresponding to eigenvalues 5 and 4, respectively. Thus, from the previous theorem, we obtain

$$P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}.$$

One can easily verify that $P^{-1}AP = D$. Observe how P is invertible.

*Recall that $A_{n\times n}$ is invertible if its columns are linearly independent or it has nonzero determinant.

Note that P is not unique, as its entries can be reordered. For the previous example, we could also have

$$P = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}.$$

Also, since we are dealing with eigenvectors, we could also have scaled versions of the columns

$$P = \begin{pmatrix} 2 & 2 \\ 3 & 6 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}.$$

Again, one can verify that $P^{-1}AP = D$. Let's do an example with a larger matrix.

Example 78. Consider the matrix

$$A = \begin{pmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{pmatrix}.$$

This matrix has the characteristic polynomial

$$C_A(x) = (x-1)(x+2)^2.$$

Thus, it has eigenvalues 1, -2, and -2. For $\lambda = 1$

$$E_1 = \text{Span}\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}.$$

For $\lambda = -2$

$$E_{-2} = \text{Span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Thus, we have

$$P = \left(\begin{array}{c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{array} \right) = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

and also

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

One can easily check that P is invertible and that $P^{-1}AP = D$.

What if two matrices have the same eigenvalues?! If one is diagonalizable, does it imply the other is? As we will show in the next example, this is not always true.

Example 79. Let

$$A = \begin{pmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{pmatrix},$$

with characteristic polynomial

$$C_A(x) = (x-1)(x+2)^2.$$

For $\lambda = 1$, we have

$$E_1 = \text{Span}\left\{\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}\right\}.$$

For $\lambda = -2$, we have

$$E_{-2} = \text{Span}\left\{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}\right\}.$$

We can construct P by repeating $\mathbf{v}_2$ twice, i.e.,

$$P = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

We still have that $AP = PD$, however, the matrix P is not invertible. We can conclude that since it is not possible to find three linearly independent eigenvectors, A is not diagonalizable.

Remark 7. If a matrix has two identical columns/rows, it has zero determinant.

Let's go over some useful results.

Theorem. If $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_r$, then $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly independent.

One corollary of this is the following

Corollary. If $A_{n\times n}$ has n distinct eigenvalues, then it is diagonalizable.

Example 80. Suppose that

$$A = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}.$$

As it is a triangular matrix, we have that its eigenvalues are $1, 2$. Since the number of distinct eigenvalues is equal to the dimension of the matrix, it is diagonalizable.

3.2.2 Application of Diagonalizability

One nice thing about diagonal matrices is that raising them to a certain power k only requires raising their elements to that power, i.e.,

$$\begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_k \end{pmatrix}^k = \begin{pmatrix} \lambda_1^k & & & \\ & \lambda_2^k & & \\ & & \ddots & \\ & & & \lambda_k^k \end{pmatrix}$$

This is because if A is diagonalizable,

$$P^{-1}AP = D \iff A = PDP^{-1}.$$

This means that

$$A^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}.$$

We can use induction to show that

$$A^k = PD^kP^{-1}.$$

Example 81. Let

$$A = \begin{pmatrix} 7 & 2 \\ -4 & 1 \end{pmatrix}.$$

To find A^k, we must first diagonalize A. Suppose we have that

$$P = \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix}.$$

In order to find A^k

$$\begin{aligned} A &= PDP^{-1} \\ A^k &= \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 5^k & 0 \\ 0 & 3^k \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2\cdot 5^k - 3^k & 5^k - 3^k \\ -2\cdot 5^k + 2\cdot 3^k & -5^k + 2\cdot 3^k \end{pmatrix}. \end{aligned}$$

3.2.3 Algebraic and Geometric Multiplicity

Definition (Algebraic/Geometric Multiplicity). Let λ be an eigenvalue of A. The *algebraic multiplicity* of λ is the number of times λ is repeated as a root of $C_A(x)$. The *geometric* multiplicity of λ is the dimension of the λ-eigenspace.

Example 82. Suppose that we are given

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad C_A(x) = x^2 - 2x + 1 = (x-1)^2.$$

We have the eigenvalue $\lambda = 1$ with algebraic multiplicity 2. The RREF of $A - I$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Since $\dim(E_1) = \text{nullity}\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) = 1$, the geometric multiplicity of $\lambda = 1$ is 1.

Theorem. If λ is an eigenvalue of A, then

$$\text{geo}(\lambda) \leq \text{alg}(\lambda).$$

Example 83. Suppose that $A_{7\times 7}$ has the characteristic equation

$$C_A(x) = (x-3)^2(x-5)(x-8)^4.$$

From this, we see that

Eigenvalues	Algebraic Multiplicity	Geometric Multiplicity
3	2	≤ 2
5	1	1
8	4	≤ 4

We are now ready to discuss the main theorem related to diagonalizability.

Theorem (Criteria for Diagonalizability). Let $A_{n\times n}$ with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$, where $p \leq n$. The following statements are equivalent.

(1) The matrix A is diagonalizable.

(2) $\sum_{i=1}^{p} \text{geo}(\lambda_i) = n$.

(3) For each i, $\text{geo}(\lambda_i) = \text{alg}(\lambda_i)$. Furthermore, if A is diagonalizable and B_i is a basis for the λ_i-eigenspace

$$B = \bigcup_{i=1}^{p} B_i$$

contains n linearly independent eigenvectors.

Let's demonstrate an example of how this is useful for large matrices using Sage.

Example 84. Suppose that

$$A = \begin{pmatrix} 9 & -16 & 16 & -40 & 24 & -40 \\ 14 & -1 & 2 & -44 & -36 & -18 \\ 8 & -6 & 7 & -30 & -6 & -20 \\ 0 & -6 & 6 & -5 & 18 & -12 \\ 2 & 0 & 0 & -6 & -5 & -2 \\ 0 & 2 & -2 & 2 & -6 & 5 \end{pmatrix}.$$

In Sage, we can determine the characteristic polynomial

$$x^6 - 10x^5 + 39x^4 - 76x^3 + 79x^2 - 42x + 9 = (x-3)^2(x-4)^4$$

DO LATER

Let's do another example to check understanding.

Example 85. Let $A_{4\times 4}$ be a matrix with eigenvalues $5, 3, -2$. Suppose that the $\lambda = 3$ eigenspace is 2-dimensional. Is A diagonalizable? Let's calculate the algebraic and geometric multiplicities of each eigenvector. We know that $\lambda = 3$ has geometric multiplicity 2 and that the other eigenvalues have geometric multiplicity at least 1. Since the sum of geometric multiplicities must be $n = 4$, the other geometric multiplicities must be 1.

Eigenvalue	Algebraic Multiplicity	Geometric Multiplicity
5	?	1
3	?	2
-2	?	1

For the algebraic multiplicities, $\lambda = 3$ must have algebraic multiplicity 2 since its eigenspace

is 2-dimensional. Since the geometric multiplicity is less than or equal to the algebraic multiplicity, we have that

Eigenvalue	Algebraic Multiplicity	Geometric Multiplicity
5	1	1
3	2	2
-2	1	1

Therefore, A is diagonalizable.

We will next do a past exam question.

Example 86. Let $A_{2\times 2}$ with eigenvalues 2 and 5.

(a) Is A diagonalizable?

(b) Must A be $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$?

Since A has $2 = n$ distinct eigenvalues, it is diagonalizable. For (b), A could also be $\left(\begin{smallmatrix} 5 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, or some other triangular matrix with the eigenvalues in the diagonal.

Example 87. Suppose that

$$A = \begin{pmatrix} 135 & a & 114 \\ -57 & -93 & b \\ -57 & -114 & -36 \end{pmatrix}$$

has the eigenvalue $\lambda = 21$ with algebraic multiplicity 2. We know that the eigenvalues of A are $21, 21, \lambda$. From last week, we know that the trace of a matrix is the sum of its eigenvalues. We have that

$$\begin{aligned} \operatorname{tr}(A) &= 21 + 21 + \lambda \\ 6 &= 42 + \lambda \\ \lambda &= 6 - 42 \\ &= -36. \end{aligned}$$

Chapter 4

Inner Product Spaces

4.1 Inner Product Spaces

In this section, we will discuss inner product spaces over the real numbers.

Definition (Inner Product). Let V be a vector space over $\mathbb{R}$. An *inner product* is a binary function $\langle , \rangle : V \times V \to \mathbb{R}$ satisfying the following properties

1) *Symmetric property*
$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle, \quad \forall \mathbf{u}, \mathbf{v} \in V.$$

2) *Additive for first coordinate*
$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v} + \mathbf{w} \rangle, \quad \forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V.$$

3) *Scalar multiplication for first coordinate.*
$$\langle k\mathbf{u}, \mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle.$$

4) *Positive definite axiom*
$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0, \quad \langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = 0.$$

A vector space V equipped with an inner product is called an *inner product space*.

Example 88. Consider $\mathbb{R}^n$. We will define a binary function $\langle , \rangle : \mathbb{R}^n \to \mathbb{R}$ by

$$\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

Checking the axioms of an inner product

1) Trivially true.

2) Also true.

3) Trivially true.

4) A bit less trivial. Let's check. We have that

$$\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle = x_1^2 + x_2^2 + \cdots + x_n^2 \geq 0.$$

We have that

$$\begin{aligned} \text{with equality} &\iff \forall x_i = 0 \\ &\iff \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{0} \end{aligned}$$

This function is actually the *dot product*.

Let's introduce some more properties of inner products.

Lemma 17. Let V be an inner product space over $\mathbb{R}$. For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $k \in \mathbb{R}$, we have

1) $\langle \mathbf{v}, \mathbf{0} \rangle = 0$.
2) $\langle \mathbf{u}, k\mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$.
3) $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.

Proof.

1) We have that

$$\begin{aligned} \langle \mathbf{v}, \mathbf{0} \rangle &= \langle \mathbf{v}, \mathbf{0} + \mathbf{0} \rangle \\ &= \langle \mathbf{v}, \mathbf{0} \rangle + \langle \mathbf{v}, \mathbf{0} \rangle. \end{aligned}$$

Therefore, we must have that $\langle \mathbf{v}, \mathbf{0} \rangle = 0$.

2)

$$\begin{aligned} \langle \mathbf{u}, k\mathbf{v} \rangle &= \langle k\mathbf{v}, \mathbf{u} \rangle \\ &= k\langle \mathbf{v}, \mathbf{u} \rangle \\ &= k\langle \mathbf{u}, \mathbf{v} \rangle \end{aligned}$$

3) LAETR

□

Let's do an example.

Example 89. Suppose that $\langle \mathbf{u}, \mathbf{u} \rangle = 1$, $\langle \mathbf{v}, \mathbf{v} \rangle = 2$, and $\langle \mathbf{u}, \mathbf{v} \rangle = -1$. Determine

$$\langle 2\mathbf{u} - \mathbf{v}, 3\mathbf{u} + 2\mathbf{v} \rangle.$$

We have that

$$\begin{aligned}
\langle 2\mathbf{u}-\mathbf{v}, 3\mathbf{u}+2\mathbf{v}\rangle &= \langle 2\mathbf{u}-\mathbf{v}, 3\mathbf{u}\rangle + \langle 2\mathbf{u}-\mathbf{v}, 2\mathbf{v}\rangle \\
&= \langle 2\mathbf{u}, 3\mathbf{u}\rangle - \langle \mathbf{v}, 3\mathbf{u}\rangle + \langle 2\mathbf{u}, 2\mathbf{v}\rangle - \langle \mathbf{v}, 2\mathbf{v}\rangle \\
&= 6\langle \mathbf{u}, \mathbf{u}\rangle - 3\langle \mathbf{v}, \mathbf{u}\rangle + 4\langle \mathbf{u}, \mathbf{v}\rangle - 2\langle \mathbf{v}, \mathbf{v}\rangle \\
&= 6\langle \mathbf{u}, \mathbf{u}\rangle + \langle \mathbf{u}, \mathbf{v}\rangle - 2\langle \mathbf{v}, \mathbf{v}\rangle \\
&= 6\cdot 1 + (-1) - 2\cdot(2) \\
&= 6 - 1 - 4 \\
&= 1.
\end{aligned}$$

Note that informally we can "multiply"

$$(2\mathbf{u}-\mathbf{v})(3\mathbf{u}+2\mathbf{v}) = 6\mathbf{uu} + \mathbf{uv} - 2\mathbf{vv}.$$

Example 90. Previously, we showed that the *dot product*, more formally known as the *Euclidean inner product*, is an inner product.

$$\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = x_1y_1 + x_2y_2 + \cdots + x_ny_n.$$

We can tweak this definition a bit to obtain the *weighted Euclidean inner product*. Let $w_1, \ldots, w_n$ be positive real numbers. We define the inner product

$$\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = w_1x_1y_1 + w_2x_2y_2 + \cdots + w_nx_ny_n.$$

to be the weighted Euclidean inner product. Note that the Euclidean inner product is merely a special case of the weighted Euclidean inner product where $w_i = 1$ for $i = 1, 2, \ldots, n$. Again, the first three axioms are fairly straightforward to prove, so we will focus our attention on the positive definite axiom. By the definition of our inner product, we have

$$\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle = w_1x_1^2 + w_2x_2^2 + \cdots + w_nx_n^2.$$

Since $w_i > 0$ and $x_i^2 \geq 0$ for all i, this summation is also greater than or equal to zero. Since all weights are strictly positive, we have that

$$\begin{aligned}
\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle = 0 &\iff w_ix_i^2 = 0, \quad \forall i \\
&\iff x_i^2 = 0, \quad \forall i \\
&\iff x_i = 0, \quad \forall i
\end{aligned}$$

Let's give an example for proving that something is *not* an inner product.

Example 91. Define

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1y_1 + 0x_2y_2.$$

This is not an inner product, as

$$\begin{aligned} \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle &= 2 \cdot 0 \cdot 0 + 0 \\ &= 0, \end{aligned}$$

however, we have that $\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right) \neq \mathbf{0}$.

Example 92. Define

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1y_1 + 3x_2y_2.$$

This is an inner product on $\mathbb{R}^2$.

There are actually infinitely many inner products that we can define on $\mathbb{R}^n$, which can be obtained by modifying weights.

Example 93. Using the definition

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1y_1 + 3x_2y_2.$$

Let's calculate

$$\left\langle \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \end{pmatrix} \right\rangle.$$

We have that

$$\begin{aligned} \left\langle \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \end{pmatrix} \right\rangle &= 2(-2)(3) + 3(1)(5) \\ &= -12 + 15 \\ &= 3. \end{aligned}$$

Do note that the default inner product is the dot product.

Remark 8. By convention, if the inner product is not explicitly given for $\mathbb{R}^n$, we default to the Euclidean inner product/dot product.

We can also have inner products for vector spaces other than $\mathbb{R}^n$ as well. Consider the following example for inner product on matrix spaces

Example 94. Suppose we have the vector space $V = M_{m\times n}(\mathbb{R})$. Define

$$\langle A, B\rangle = \operatorname{tr}\left(AB^T\right), \quad A, B \in M_{m\times n}.$$

Let's see if this satisfies the axioms of a inner product

1) For symmetry, we have that

$$\begin{aligned}\langle B,A\rangle &= \operatorname{tr}\left(BA^T\right) \\ &= \operatorname{tr}\left((BA^T)^T\right) && (\operatorname{tr}(X) = \operatorname{tr}(X^T)) \\ &= \operatorname{tr}\left(AB^T\right) && ((AB)^T = B^T A^T) \\ &= \langle A,B\rangle\end{aligned}$$

2) Pretty straightforward.

3) Pretty straightforward.

4) A little tricky. We want to show that $\langle A,A\rangle \geq 0$ for all A and that the only case that $\langle A,A\rangle = 0$ is when $A = \mathbf{0}$. We have that

$$A = \begin{pmatrix} \dots r_1 \dots \\ \dots r_2 \dots \\ \vdots \\ \dots r_n \dots \end{pmatrix}$$

$$A^T = \begin{pmatrix} \vdots & \vdots & & \\ \vdots & \vdots & & \\ r_1 & r_2 & \dots & \dots \\ \vdots & \vdots & & \end{pmatrix}.$$

Furthermore, observe that entry (i,j) in AA^T is $\vec{r_i}\cdot\vec{r_j}$ and entry (i,i) in AA^T is $\vec{r_i}\cdot\vec{r_i} \geq 0$. We have that

$$\operatorname{tr}\left(AA^T\right) = \sum_i (\vec{r_i}\cdot\vec{r_i}) \geq 0.$$

Moreover

$$\begin{aligned}\operatorname{tr}\left(AA^T\right) = 0 &\iff \vec{r_i}\cdot\vec{r_i} = 0 \\ &\iff \vec{r_i} = \mathbf{0}, \quad \forall i \\ &\iff A = 0\end{aligned}$$

Let's do a calculated based example with the inner product from the previous example.

Example 95. Suppose we are working with the vector space $V = M_{2\times 2}(\mathbb{R})$. We have that for

$$\langle A,B\rangle = \operatorname{tr}\left(AB^T\right)$$

We can calculate things like

$$\begin{aligned}\left\langle \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ 2 & 2 \end{pmatrix} \right\rangle &= \operatorname{tr}\left\{ \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 1 & 2 \end{pmatrix} \right\} \\ &= \operatorname{tr}\left\{ \begin{pmatrix} 1 & 7 \\ 1 & 2 \end{pmatrix} \right\} \\ &= 3.\end{aligned}$$

How about inner product on the polynomial space?

Example 96. Suppose that $V = \mathbf{P}_n(x)$. Define

$$\begin{aligned} p(x) &= a_0 + a_1 x + \cdots + a_n x^n \\ q(x) &= b_0 + b_1 x + \cdots + b_n x^n. \end{aligned}$$

We will define the inner product

$$\langle p, q \rangle = a_0 b_0 + a_1 b_1 + \cdots + a_n b_n.$$

We can show that this is an inner product. It can also be weighted. As an example, according to this definition, let's calculate

$$\begin{aligned} \langle 1+2x, 3-5x \rangle &= (1)(3) + (2)(-5) \\ &= 3 - 10 \\ &= -7. \end{aligned}$$

Example 97. Another inner product on the polynomial space is the *evaluation inner product*. Consider $p(x), q(x) \in \mathbf{P}_n(x)$. If $x_0, x_1, x_2, \ldots, x_n$ are distinct real numbers, then

$$\langle p, q \rangle = p(x_0)q(x_0) + p(x_1)q(x_1) + \cdots + p(x_n)q(x_n).$$

This is an inner product. Why? Let's verify the axioms. Linear properties (scalar multiplication and addition to first coordinate) are pretty obvious, as is symmetry. Let's verify the positive definite axiom.

$$\langle p, p \rangle = p(x_0)^2 + p(x_1)^2 + \cdots + p(x_n)^2.$$

As $\langle p, p \rangle$ is the sum of squared real numbers, it is greater than or equal to zero. If $\langle p, p \rangle = 0$, it must be the case that $p(x_i) = 0$ for $i = 1, 2, \ldots, n$. Why does this mean that $p = \mathbf{0}$, the zero-polynomial? Assume to the contrary that it is not. That would imply that $x_0, x_1, \ldots, x_n$ are all roots of p. However, p has degree at most n, making it impossible to have $n+1$ distinct roots. This is a contradiction to the Fundamental Theorem of Algebra.

Example 98. In $\mathbf{P}_2(x)$. Suppose that

$$x_0 = -2, \quad x_1 = 0, \quad x_2 = 2.$$

We shall take $p(x) = x^2$ and $q(x) = x + 1$. Then

$$\begin{aligned} \langle p, q \rangle &= p(-2)q(-2) + p(0)q(0) + p(2)q(2) \\ &= (4)(-1) + (0)(1) + (4)(3) \\ &= -4 + 0 + 12 \\ &= 8. \end{aligned}$$

Observe that this is equivalent to

$$\begin{pmatrix} p(x_0) \\ p(x_1) \\ p(x_2) \end{pmatrix} \cdot \begin{pmatrix} q(x_0) \\ q(x_1) \\ q(x_2) \end{pmatrix}.$$

4.1.1 Inner Products on Function Spaces

Suppose that $V = C[a,b]$, the vector space of continuous functions defined on $[a,b]$. We define the inner product

$$\langle f, g\rangle = \int_a^b f(x)g(x)\,dx$$

The linearity related properties are quite easy to show using properties of definite integrals. For positive-definite, it requires some real analysis but note that

$$\langle f, f\rangle = \int_a^b f(x)^2\,dx \geq 0, \quad \int_a^b f(x)^2\,dx = 0 \implies f \mapsto 0$$

Example 99. We will work with $V = C[-1,1]$. We will have the inner product

$$\langle f, g\rangle = \int_{-1}^1 f(x)g(x)\,dx.$$

Suppose $f = x$ and $g = x^2$. We would have

$$\begin{aligned}\langle x, x^2\rangle &= \int_{-1}^1 x^3\,dx \\ &= \left.\frac{x^4}{4}\right|_{-1}^1 \\ &= \frac{1}{4} - \frac{1}{4} \\ &= 0.\end{aligned}$$

In the language of inner product spaces, we say that f and g are *orthogonal*, as their inner product is zero.

4.1.2 Norm and Distance

With inner products, we can define the concepts of norm and distance.

Definition (Norm/Distance). If V is a real inner product space, then the *norm* (length) of a vector $\mathbf{v} \in V$ is define to be

$$\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}.$$

The *distance* between two vectors $\mathbf{u}$ and $\mathbf{v}$ is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Note that in $\mathbb{R}^n$

$$\left\|\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right\| = \sqrt{x_1^2 + \cdots + x_n^2}$$

Theorem (Properties of Norm).

1) $\|\mathbf{v}\| \geq 0$ with equality if and only if $\mathbf{v} = \mathbf{0}$.

2) $\|k\mathbf{v}\| = |k|\|\mathbf{v}\|$

3) $d(\mathbf{u},\mathbf{v}) = d(\mathbf{v},\mathbf{u})$

4) $d(\mathbf{u},\mathbf{v}) \geq 0$ with equality if and only if $\mathbf{u} = \mathbf{v}$

Proof.

1. Positive definite from inner product
2. We have that

$$\begin{aligned}\sqrt{\langle k\mathbf{v}, k\mathbf{v}\rangle} &= \sqrt{k^2\langle \mathbf{v},\mathbf{v}\rangle} \\ &= |k|\sqrt{\langle \mathbf{v},\mathbf{v}\rangle} \\ &= |k|\|\mathbf{v}\|.\end{aligned}$$

3. Observe that

$$\begin{aligned}d(\mathbf{u},\mathbf{v}) &= \|\mathbf{u}-\mathbf{v}\| \\ &= \|-(\mathbf{v}-\mathbf{u})\| \\ &= \|\mathbf{v}-\mathbf{u}\| \\ &= d(\mathbf{v},\mathbf{u}).\end{aligned}$$

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□

With the concept of norm, we can also define a *unit vector*.

Definition (Unit Vector). $\mathbf{v} \in V$ is a unit vector if $\|\mathbf{v}\| = 1$.

We can *normalize* a vector $\mathbf{v} \neq 0$ to obtain the unit vector

$$\frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Example 100. Find the distance between $f(x) = \sin(x)$ and $g(x) = 1$ in $C[0,\pi]$, using

$$\langle f, g\rangle = \int_0^\pi f(x)g(x)\,dx.$$

We have that

$$\begin{aligned}d(\sin x, 1) &= \|\sin x - 1\| \\ &= \sqrt{\langle \sin x - 1, \sin x - 1\rangle}.\end{aligned}$$

Let's calculate the inner product of $\sin x - 1$ with itself.

$$\begin{aligned}\langle \sin x - 1, \sin x - 1\rangle &= \int_0^{\pi} (\sin x - 1)^2\, dx \\ &= \int_0^{\pi} \sin^2 x - 2\sin x + 1 \\ &= \int_0^{\pi} \frac{1-\cos 2x}{2}\, dx - \int_0^{\pi} 2\sin x\, dx + \int_0^{\pi} 1\, dx\end{aligned}$$

For the first integral, let $u = 2x$, then $du = 2dx$. Thus

$$\begin{aligned}\int_0^{2\pi} \frac{1-\cos(u)}{2}\,\frac{du}{2} &= \frac{1}{4}u\Big|_0^{2\pi} - \frac{\sin(u)}{4}\Big|_0^{2\pi} \\ &= \frac{\pi}{2}.\end{aligned}$$

For the second integral

$$\begin{aligned}\int_0^{\pi} 2\sin x\, dx &= -2\cos x \int_0^{\pi} \\ &= 2+2 \\ &= 4\end{aligned}$$

For the final term

$$\int_0^{\pi} 1\, dx = x\Big|_0^{\pi} = \pi.$$

So the final answer would be

$$d = \frac{\pi}{2} - 4 + \pi = \frac{3\pi}{2} - 4.$$

Remark 9. Some trig identities to remember

1. $$\sin^2 x = \frac{1-\cos 2x}{2}$$
2. $$\cos^2 x = \frac{1+\cos 2x}{2}$$
3. $$\cos(2x) = \cos^2 x - \sin^2 x$$

Example 101. Suppose that $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ are such that

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v}\rangle &= 2 \\ \langle \mathbf{v}, \mathbf{w}\rangle &= -6 \\ \langle \mathbf{u}, \mathbf{w}\rangle &= -3.\end{aligned}$$

We also know that

$$\begin{aligned}\|\mathbf{u}\| &= 1\\ \|\mathbf{v}\| &= 2\\ \|\mathbf{w}\| &= 7.\end{aligned}$$

Find $\langle 2\mathbf{v}-\mathbf{w}, 3\mathbf{u}+2\mathbf{w}\rangle$. By linear properties, we can kind of distribute multiplicatively

$$\begin{aligned}\langle 2\mathbf{v}-\mathbf{w}, 3\mathbf{u}+2\mathbf{w}\rangle &= 6\langle \mathbf{v},\mathbf{u}\rangle + 4\langle \mathbf{v},\mathbf{w}\rangle - 3\langle \mathbf{w},\mathbf{u}\rangle - 2\langle \mathbf{w},\mathbf{w}\rangle\\ &= 6\cdot 2 + 4\cdot -6 - 3\cdot -3 - 2\langle \mathbf{ww}\rangle\\ &= -3 - 2\langle \mathbf{w},\mathbf{w}\rangle.\end{aligned}$$

As we are given that $\|\mathbf{w}\| = \sqrt{\langle \mathbf{w},\mathbf{w}\rangle}$, we have $\langle \mathbf{w},\mathbf{w}\rangle = (\|\mathbf{w}\|)^2 = 49$. Thus

$$\begin{aligned}\langle 2\mathbf{v}-\mathbf{w}, 3\mathbf{u}+2\mathbf{w}\rangle &= -3 - 2\langle \mathbf{w},\mathbf{w}\rangle\\ &= -3 - 2\cdot 49\\ &= -101.\end{aligned}$$

What if we also wanted to find $\|\mathbf{u}+\mathbf{v}\|$?

$$\begin{aligned}\|\mathbf{u}+\mathbf{v}\| &= \sqrt{\mathbf{u}+\mathbf{v},\mathbf{u}+\mathbf{v}}\\ &= \sqrt{\langle \mathbf{u},\mathbf{u}\rangle + 2\langle \mathbf{u},\mathbf{v}\rangle + \langle \mathbf{v},\mathbf{v}\rangle}\\ &= \sqrt{1+2\cdot 2+4}\\ &= \sqrt{9}\\ &= 3.\end{aligned}$$

We can even write the trick we used as a theorem.

Theorem. $\|\mathbf{v}\|^2 = \langle \mathbf{v},\mathbf{v}\rangle$.

Chapter 5

Extra: Applications of Linear Algebra

5.1 Translations in $\mathbb{R}^2$

A *translation* in $\mathbb{R}^2$ by a vector $\binom{a}{b}$ is a map

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}.$$

This is not a linear transformation, unless $(a,b)^T = \mathbf{0}$. In computer graphics, translations are usually understood as moving one point to another point. If we had some sort of box in Photoshop and wanted to move it, how would we do so? Let's introduce the idea of *homogeneous coordinates*. Any vector $(x,y)^T \in \mathbb{R}^2$ can be written in homogeneous coordinates

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

Now translations would look like

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} x+a \\ y+b \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

We call the matrix above the *translation matrix*. Now that we have matrix multiplication, we can easily write programs to perform these operations/calculations.

5.2 Rotation About Any Point in $\mathbb{R}^2$

Another important thing in computer graphics is rotation around a point, a transformation which is also non-linear. The idea is to translate the point to the origin, perform the rotation, then translate back to the original position. To translate a point (a,b) to the origin, we'd multiply by

$$\begin{pmatrix} 1 & 0 & -a \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{pmatrix}$$

Then we'd rotate about the origin and translate back by $\binom{a}{b}$.

5.3 Linear Recurrence Relations

We can use linear algebra to find closed forms for linear recurrence relations. Suppose we wanted to find the general formula for the sequence defined by

$$x_{n+1} = 6x_{n+1} - 8x_n, \quad x_0 = -1, x_1 = 0.$$

The first few elements of the sequence are

$$-1, 0, 8, 48, \ldots$$

To find a closed form using linear algebra, we can write

$$\mathbf{v}_n = \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix}, \quad n \geq 0.$$

We have that

$$\begin{aligned} \mathbf{v}_0 &= \begin{pmatrix} x_1 \\ x_0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \\ \mathbf{v}_1 &= \begin{pmatrix} x_2 \\ x_1 \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \end{pmatrix}. \end{aligned}$$

By definition, we have

$$\begin{aligned} \mathbf{v}_{n+1} &= \begin{pmatrix} x_{n+2} \\ x_{n+1} \end{pmatrix} \\ &= \begin{pmatrix} 6x_{n+1} - 8x_n \\ x_{n+1} \end{pmatrix} \\ &= \underbrace{\begin{pmatrix} 6 & -8 \\ 1 & 0 \end{pmatrix}}_{A} \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} \end{aligned}$$

Now we have the relationship

$$\mathbf{v}_{n+1} = A\mathbf{v}_n.$$

This means that

$$\begin{aligned} \mathbf{v}_1 &= A\mathbf{v}_0 \\ \mathbf{v}_2 &= A\mathbf{v}_1 \\ &= A^2\mathbf{v}_0 \\ \mathbf{v}_3 &= A\mathbf{v}_2 \\ &= A^3\mathbf{v}_0 \\ &\vdots \end{aligned}$$

By induction, we have that

$$\mathbf{v}_n = A^n\mathbf{v}_0.$$

We can quickly calculate A^n on a computer, we're set. We can do so by diagonalizing it with

$$P = \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

We would then have that since $P^{-1}AP = D$ or in other terms $A = PDP^{-1}$ so

$$\begin{aligned} A &= PDP^{-1} \\ A^n &= PD^nP^{-1} \\ &= \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & 4^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix}^{-1} \end{aligned}$$

If we solve this, we would have that

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = \begin{pmatrix} 4^{n+1} - 2^{n+2} \\ 4^n - 2^{n+1} \end{pmatrix}.$$

Exercise

Find closed form of F_n (Fibonacci sequence).